

Milestone 3 – Completed Functionalities with Screenshots

FOR

LICENSE & BI DASHBOARD WEB APPLICATION

SUBMITTED TO

IVTECH COMPUTER SYSTEMS LLC

Version: 1.0

13th JUNE 2025

Prepared By:

SPEC INDIA

Revision History

Date	Version	Author	Created / Updated / Reviewed / Verified / Approved By
13-06-2025	1.0	Parth Jotangiya	- Initial Draft
13-06-2025	1.0	Yaman Kavishwar	- Reviewed and approved

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1. Introduction

- This document serves as a comprehensive summary of all the functionalities that were planned, developed, and completed as part of **Milestone 3** for the License & BI Dashboard Web Application project. It includes detailed information about each feature, covering its objective, scope, and expected behavior.
- Each functionality is thoroughly described and supported with annotated screenshots to visually demonstrate successful implementation. These visual validations help in ensuring that the delivered features align with the requirements and offer clarity for future reference, testing, or client verification.

1.1. Items in the Scope

Sr. No	Functionality Title	Description	Status
1	License Web Application (<i>Partial</i>)	• Allows users to log in using their credentials. • Super users can create and manage sub-users. • Role-based access: create and manage roles and their permissions. • Manage customer and distributor details. • Generate and email licenses to distributors.	Completed
2	Camera API Understanding and Integration	• Reviewed SUN API and Wise API documentation. • Gained complete understanding of available features and functionalities as per the scope. • Focus on retrieving camera details and associated event data	Completed
3	BI Dashboard Web Application (<i>Partial</i>)	• User authentication with credential-based login. • Super users can manage sub-users. • Role and permission management. • Ability to manage camera devices and AI boxes. • Floor plan and zone management. • Camera-to-zone configuration. • Multi-site management.	Completed
4	Data Ingestion (<i>Partial</i>)	• Scheduled using cron jobs at specific time intervals. • Ability to replicate data from camera devices.	Completed

1.2. Items outside the Project Scope

- **Creation of APIs which will be directly interacted with AI camera managed by Hanwha.**
- **SPEC INDIA will only utilize camera APIs managed by Hanwha** and will replicate/ingest the necessary information in the database (if required) & display that required information on the web application.
- **Multilingual support** for web application. Application will be only in English language.
- **Responsiveness to Tablet & Mobile.** Application will be supported to desktop computer screen only.
- **The development of Mobile Application** for Android and iOS native integration will be out of scope.
- **QA Testing is out of scope** because client team will do it from their end.
- **Failover plan** for the database in case of any future failures.

2. Detailed Screenshots for Each Functionality

2.1 License Web Application (*Partial*)

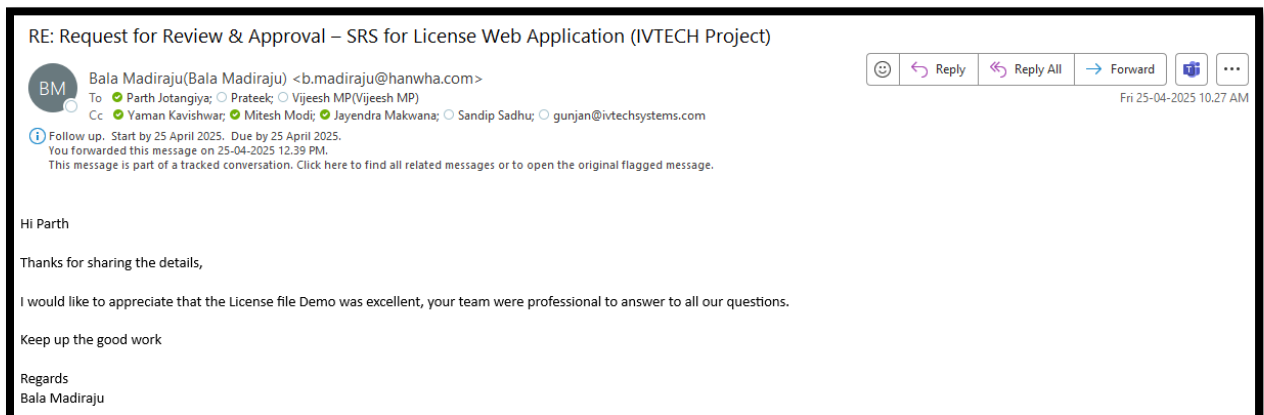
The **License Web Application** is a centralized platform designed to efficiently manage user access, licensing, and customer-distributor relationships for software license products. It allows authenticated users to securely log in, while super users can create and manage sub-users with specific roles and permissions through a flexible role-based access control system. This ensures that each user only has access to the features and data relevant to their responsibilities.

In addition, the application streamlines customer and distributor management by allowing users to store and maintain essential contact and business details. One of its key features is the ability to generate and email software licenses directly to distributors, enhancing operational efficiency and reducing manual errors. This functionality makes the system ideal for organizations seeking a secure, scalable, and user-friendly solution for license management.

Here is the link of created Figma Design for License Web Application: [Click Here](#)

IMPORTANT NOTE:

Initially, it was planned to deliver the **License Web Application** partially under **Milestone 3**. However, due to changes in the execution plan and a comprehensive flow of **requirement gathering**, all associated **features and functionalities** were finalized at the very first stage. As a result, **SPEC INDIA** successfully **developed and demonstrated the complete License Web Application** to **Bala and his team** on **Thursday, 24th April 2025 at 5:16 PM IST**. The **demonstration was well-received**, and **Bala shared his appreciation via email on Friday, 25th April 2025 at 10:27 AM IST**:



2.2 Camera API Understanding and Integration

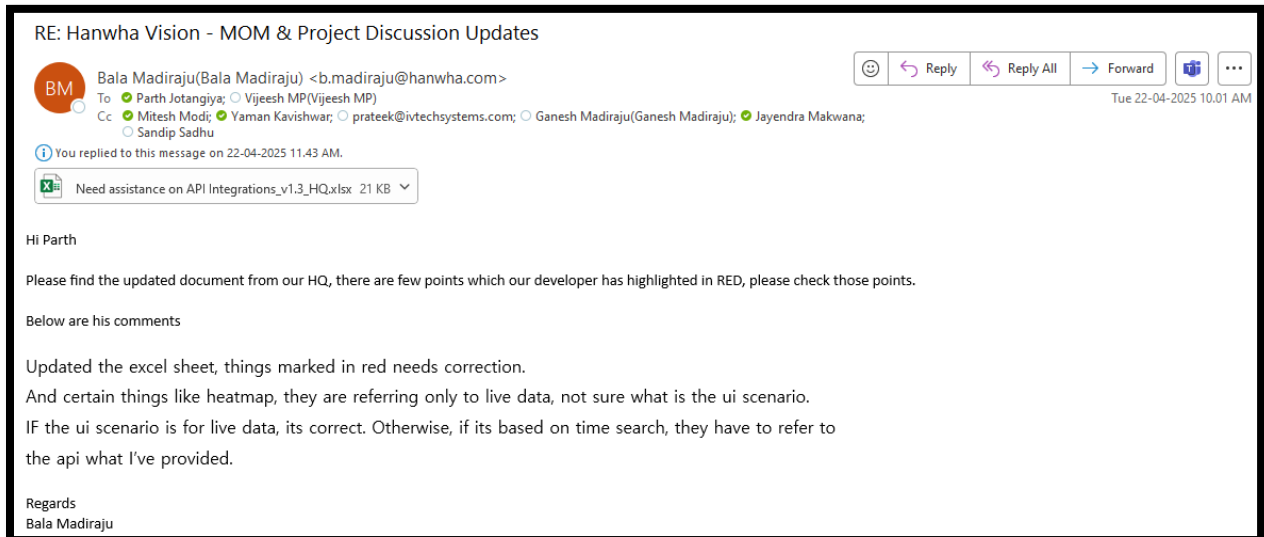
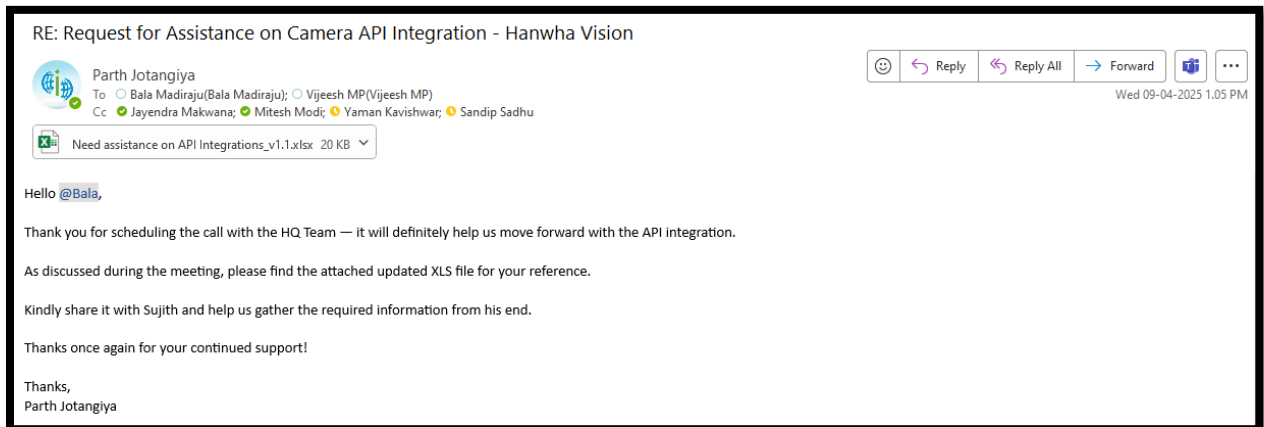
The **SPEC INDIA team** has thoroughly reviewed all the documents shared by **Bala and his team** related to the **SUNAPI and Wise APIs**. During the review, the team encountered several **uncertainties regarding the exact API endpoints** required for specific features and functionalities. To address these concerns, we reached out to **Bala**, who subsequently scheduled meetings with their **HQ Team** to obtain the necessary insights on API integration and endpoint details.

Following **multiple discussions with the HQ Team**, particularly with **Sujith**, we have successfully prepared a detailed **XLS file** outlining and finalizing the **API endpoints associated with each feature and functionality** that involves **Camera API calls**.

We continue to follow the **same practices** whenever we require any further **assistance or clarifications**.

Screenshots:

- ➔ Please find below a few **screenshots from our email conversations** highlighting the discussions around **API understanding and integration details**:



2.3 BI Dashboard Web Application (*Partial*)

The **BI Dashboard Web Application** is a platform designed with key foundational modules to support **user access control** and **device management**. It includes a secure, **credential-based user authentication system**, and a comprehensive **role and permission management** structure that allows **super users** to create and manage **sub-users** with varying levels of access based on assigned roles.

Additionally, core functionalities have been developed for **camera** and **AI box management**, enabling authorized users to add, update, or remove devices. The system features **floor plan and zone management**, allowing users to define layouts and associate specific areas with camera zones. A **camera-to-zone configuration module** has also been implemented to map devices to their respective zones. Furthermore, the application supports **multi-site management** for deployments across multiple locations. Users can **track and acknowledge generated events**, **configure system settings**, and **customize their dashboard** by adding or removing widgets as needed.

Here is the link of created Figma Design for BI Dashboard Web Application:

- Light Theme: [Click Here](#)
- Dark Theme: [Click Here](#)

IMPORTANT NOTE:

Milestone 3 includes the delivery of the following features:

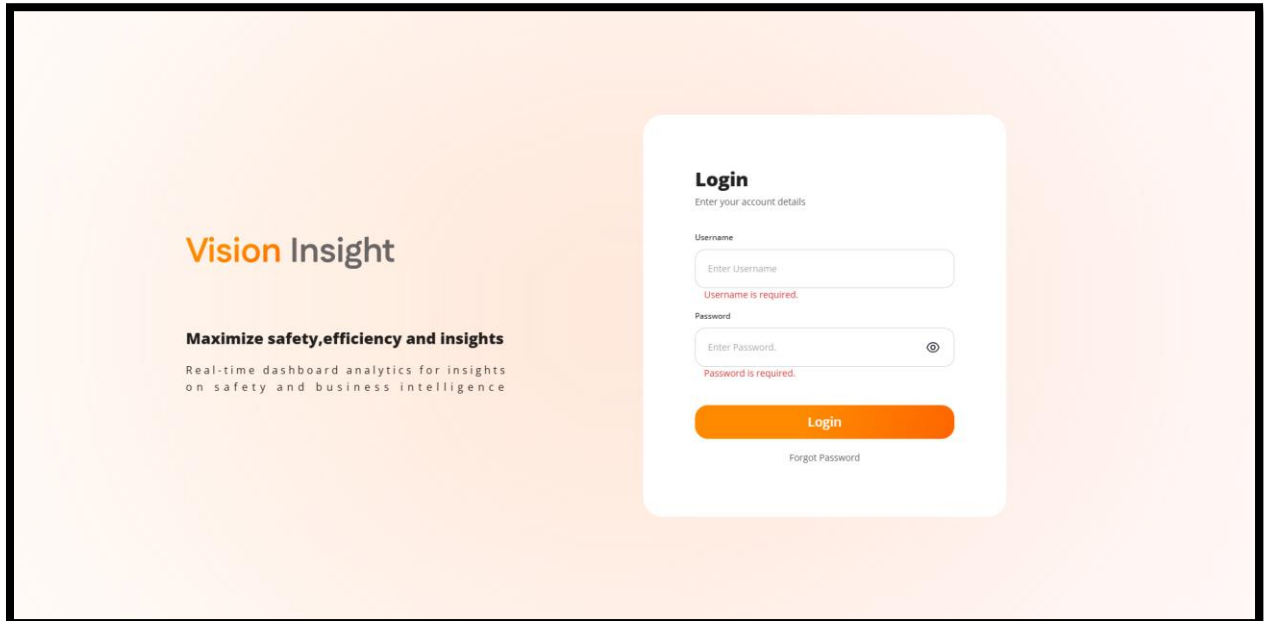
- User authentication with credential-based login.
- Super users will have the ability to manage sub-users.
- Role-based access control with permission management.
- Functionality to manage camera devices and AI boxes.
- Floor plan and zone configuration capabilities.
- Mapping of cameras to designated zones.
- Support for managing multiple sites.

1) User authentication with credential-based login

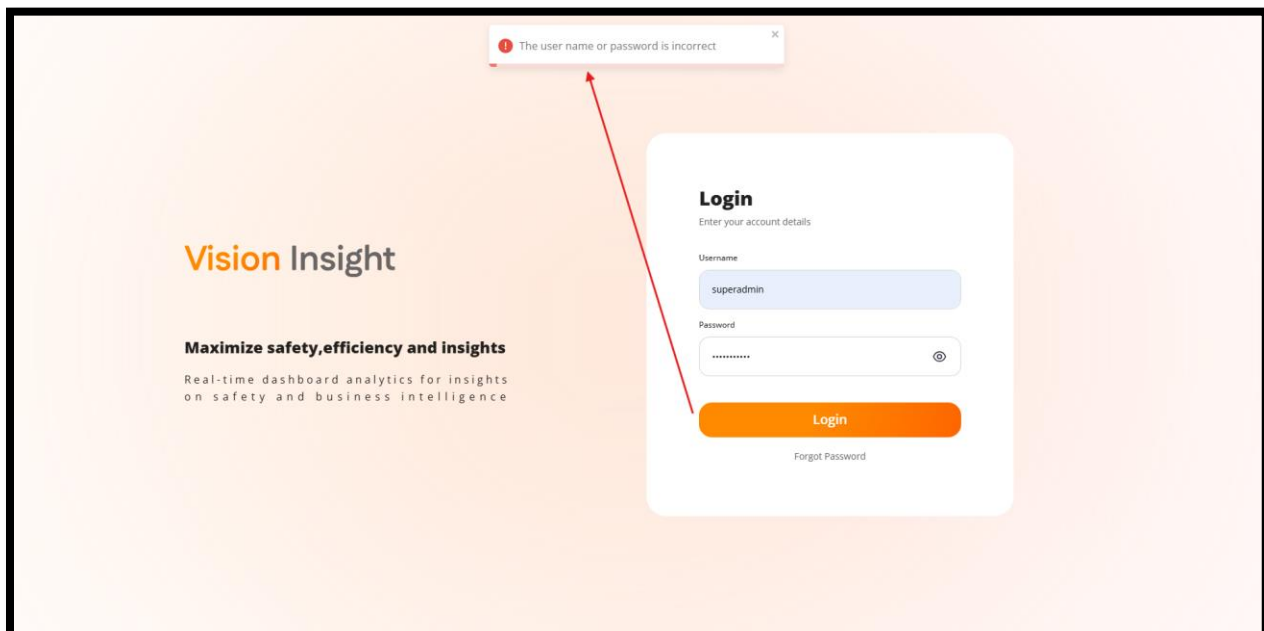
This involves implementing a secure system for users to log in using unique credentials (e.g., username/email and password).

Screenshots:

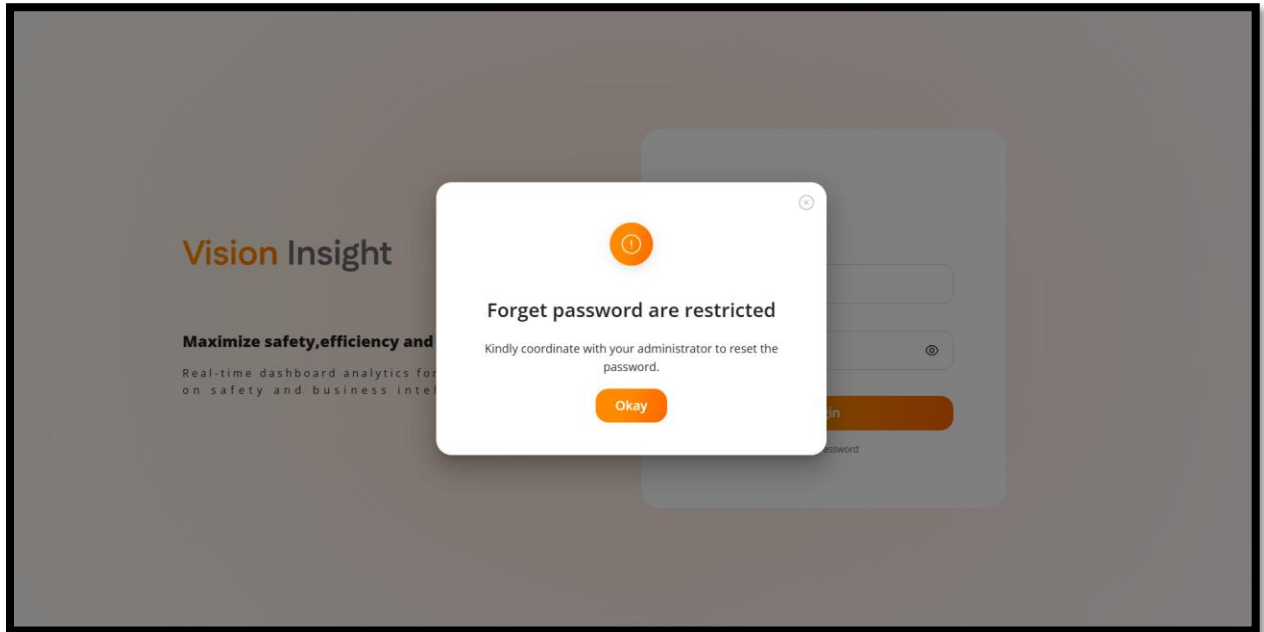
- ➔ The system will perform **field validation** when a user attempts to **log in without entering unique credential details**:



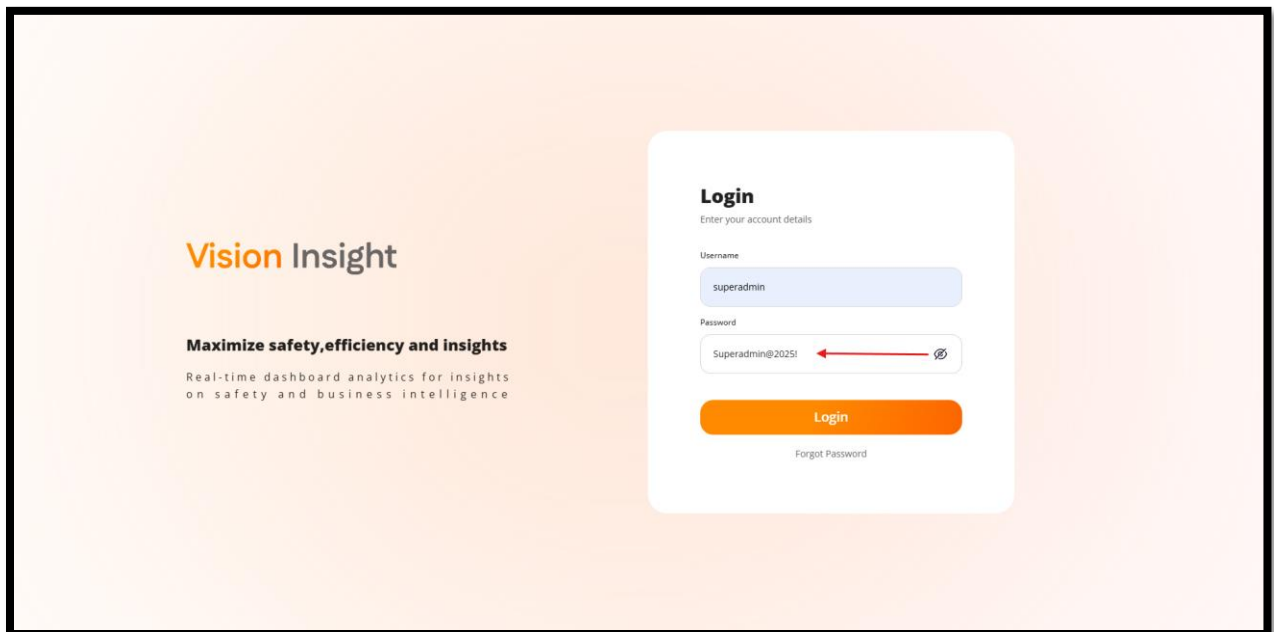
- The system will **authenticate valid credentials** when the user attempts to **log in using their unique credentials**. If the credentials are **not valid**, the system will display an **appropriate error message** to the user:



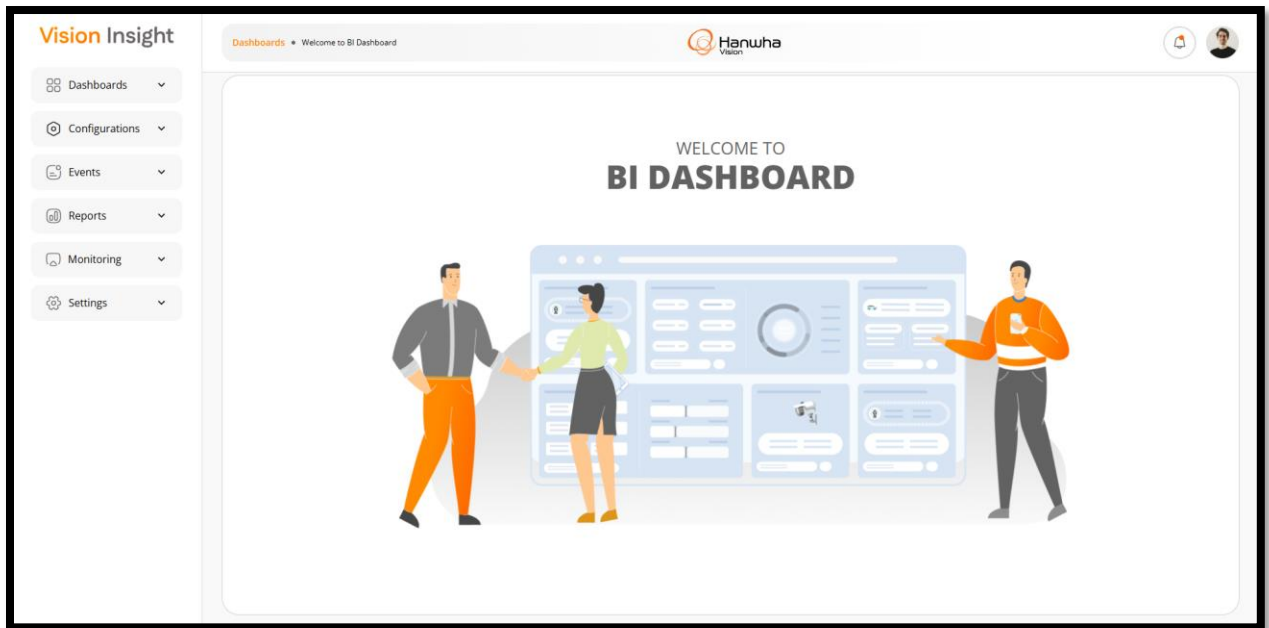
- The system will display the **following popup message** when a user attempts to use the "**Forgot Password**" option, as the system **does not allow users to reset their own passwords**:



→ The system will allow the user to **toggle the visibility** of the **entered password** by clicking on the **eye icon**, as highlighted below:



➔ After a **successful login**, users will be **redirected to the Welcome page**:

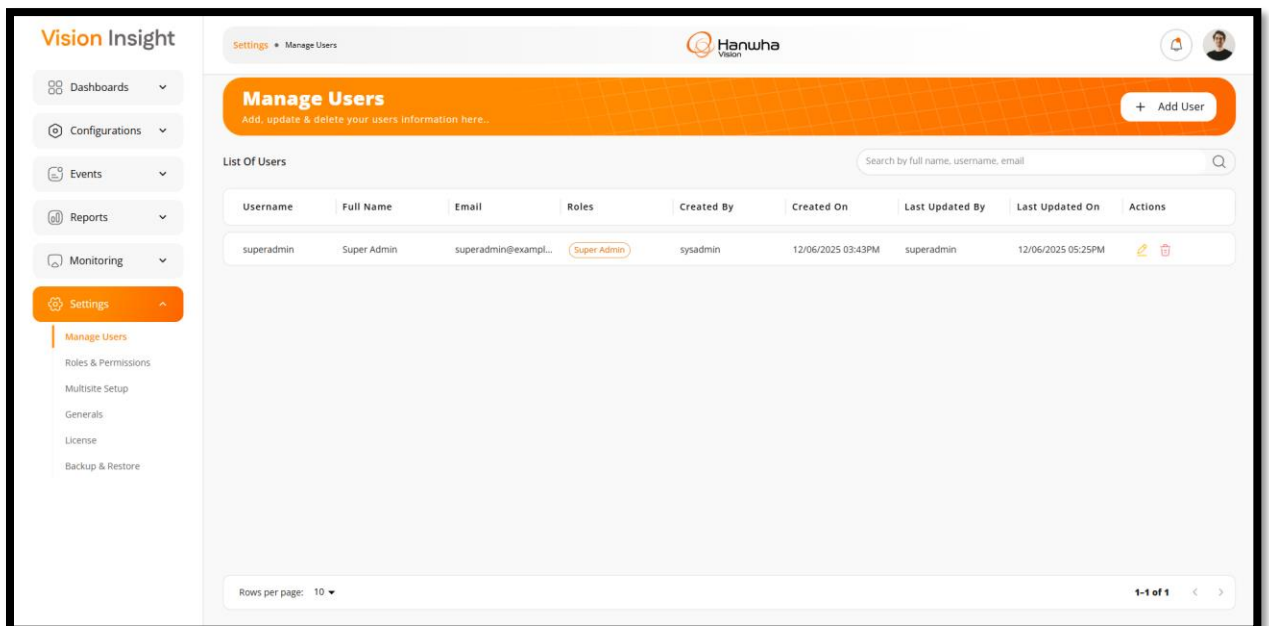


2) Super users will have the ability to manage sub-users

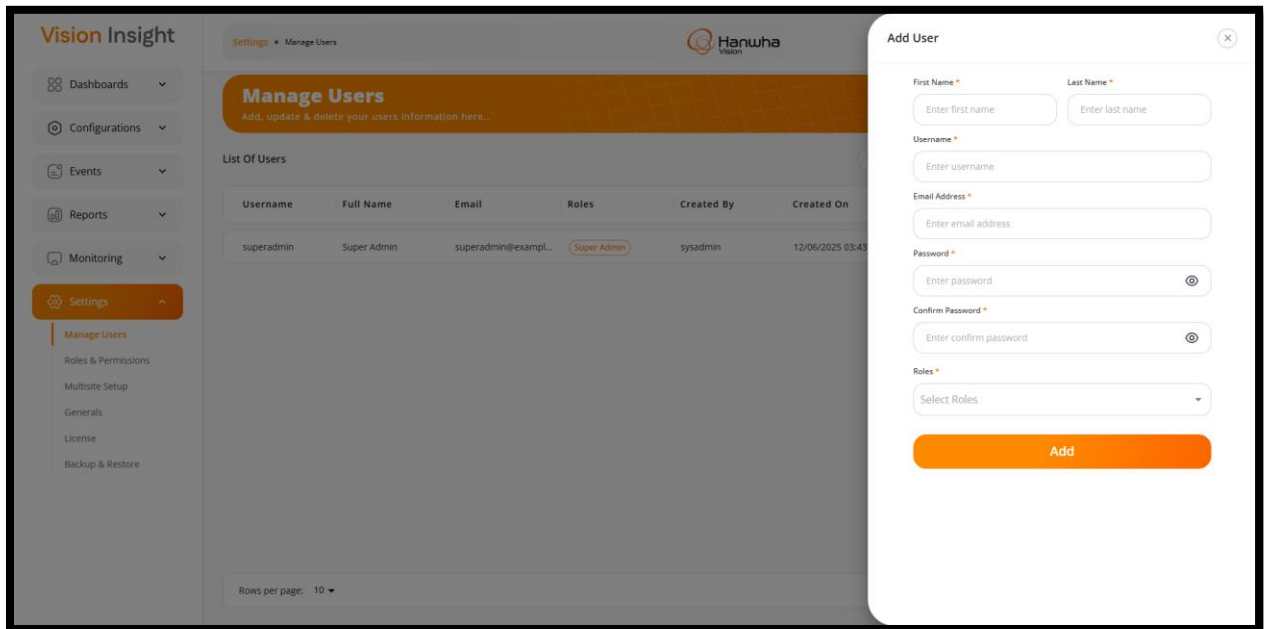
Super users will have the capability to create, modify, and delete sub-user accounts, providing hierarchical control over user access.

Screenshots:

➔ The screen below displays the **list of all created users**, along with options to **Search, Add, Edit,** and **Delete** users:



➔ The screen below represents the **form for creating a new**, allowing users to **enter relevant details** and **submit the information** for user creation:



Vision Insight | Settings • Manage Users | Hanwha Vision

Manage Users

Add, update & delete your users information here...

List Of Users

Username	Full Name	Email	Roles	Created By	Created On
superadmin	Super Admin	superadmin@exampl...	Super Admin	sysadmin	12/06/2025 03:43

Rows per page: 10

Add User

First Name *

Last Name *

Username *

Email Address *

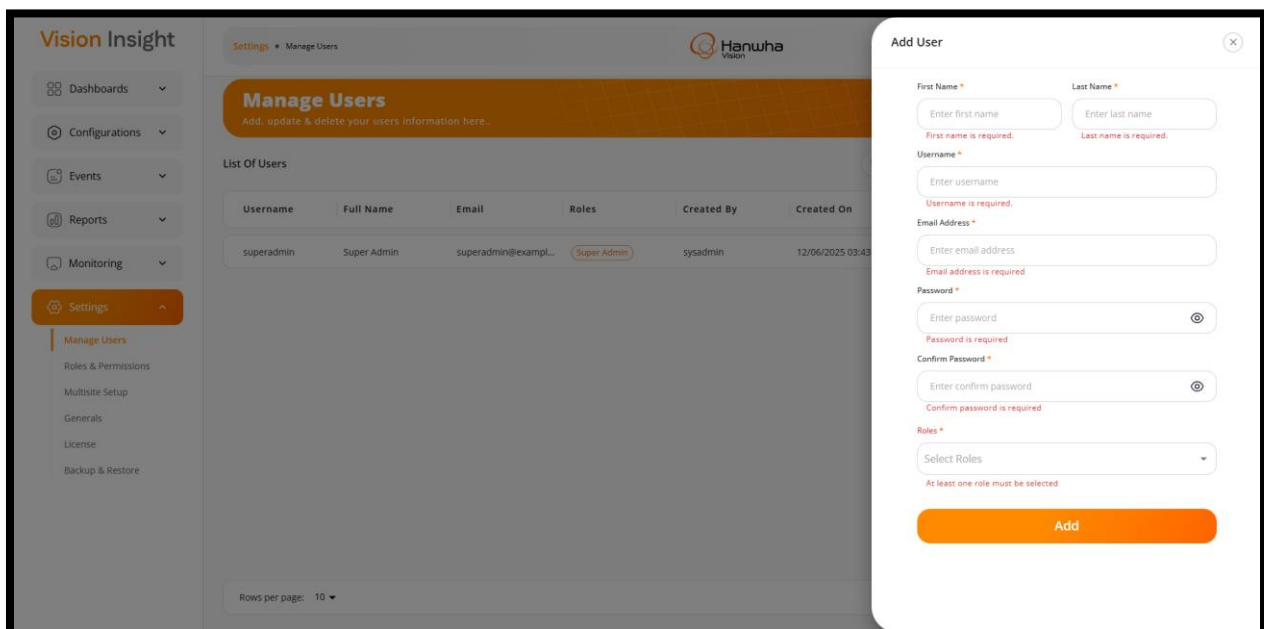
Password *

Confirm Password *

Roles *

Add

➔ The system will perform **all necessary validations** when **adding a new user** to ensure the entered information is valid and accurate:



Vision Insight | Settings • Manage Users | Hanwha Vision

Manage Users

Add, update & delete your users information here...

List Of Users

Username	Full Name	Email	Roles	Created By	Created On
superadmin	Super Admin	superadmin@exampl...	Super Admin	sysadmin	12/06/2025 03:43

Rows per page: 10

Add User

First Name *
First name is required.

Last Name *
Last name is required.

Username *
Username is required.

Email Address *
Email address is required

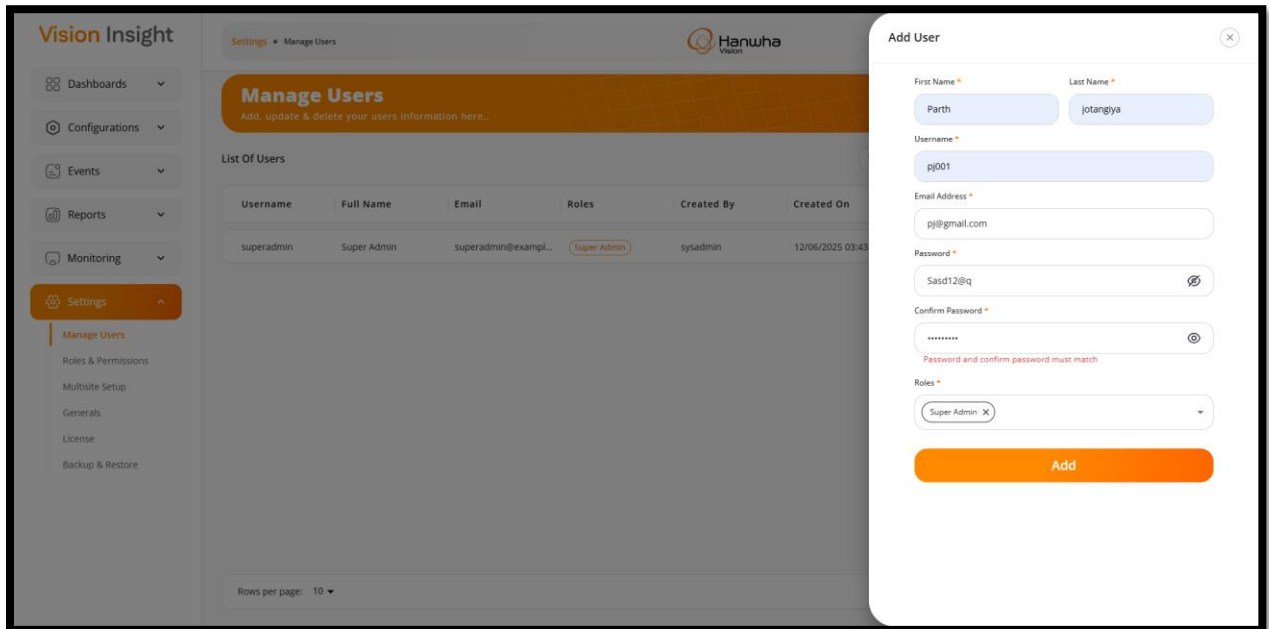
Password *
Password is required

Confirm Password *
Confirm password is required

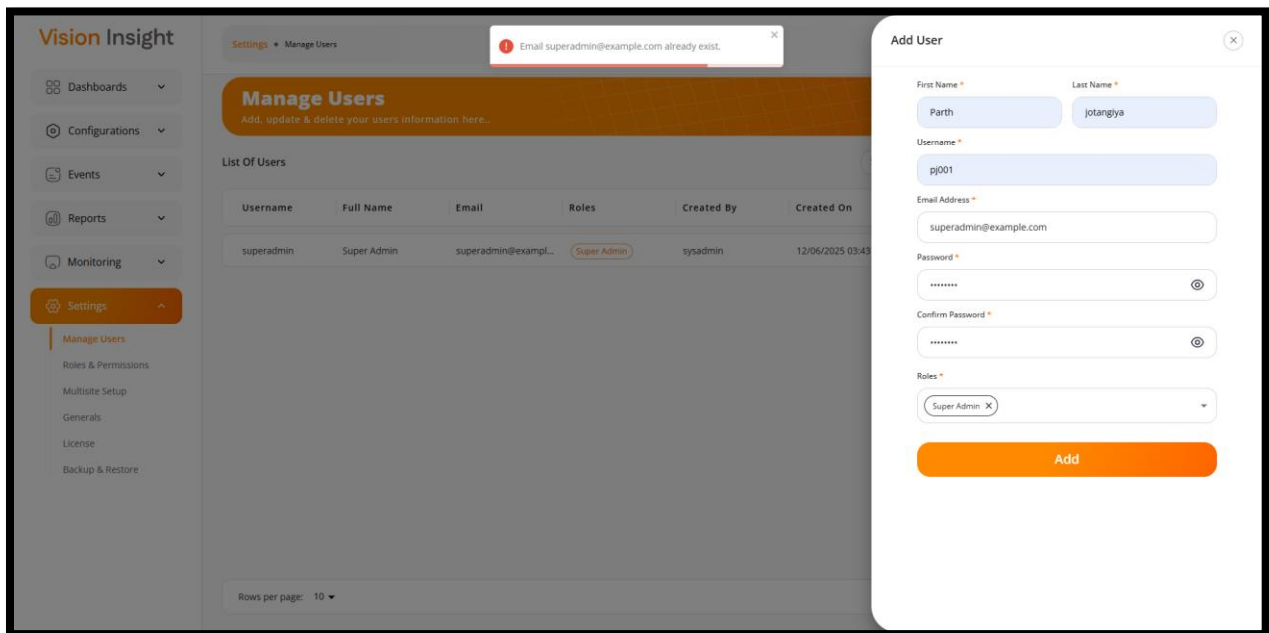
Roles *
At least one role must be selected

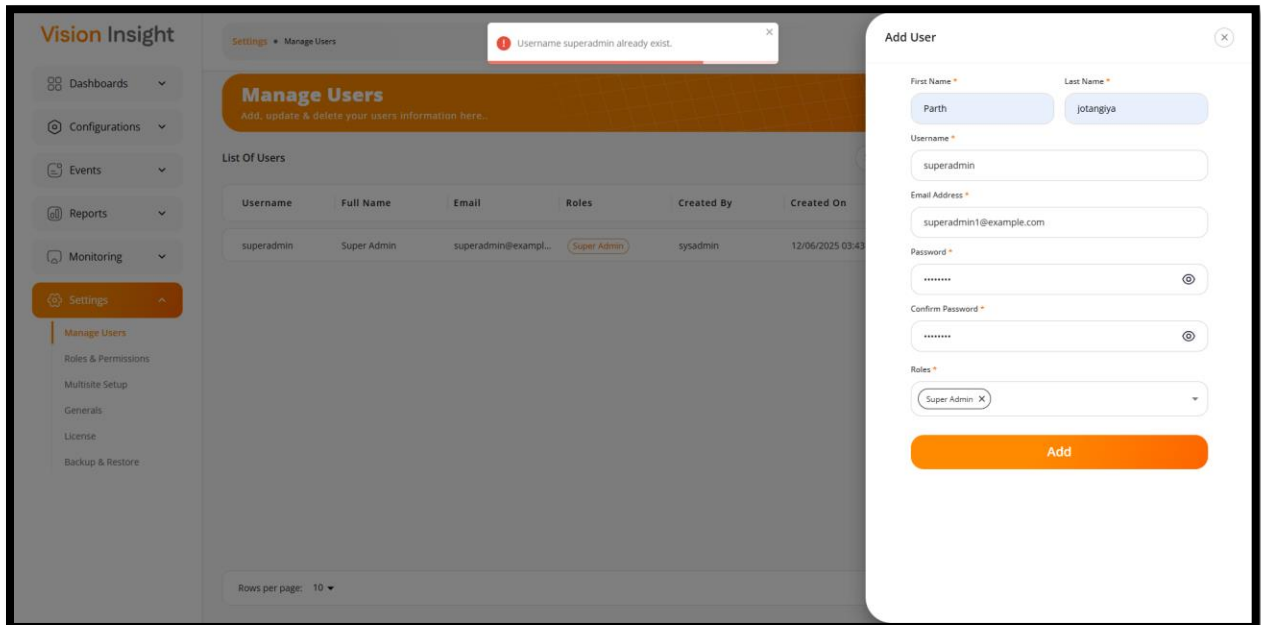
Add

➔ The system will perform **validation on both the Password and Confirm Password fields**. If the values **do not match**, the system will **display an appropriate error message** to the user:

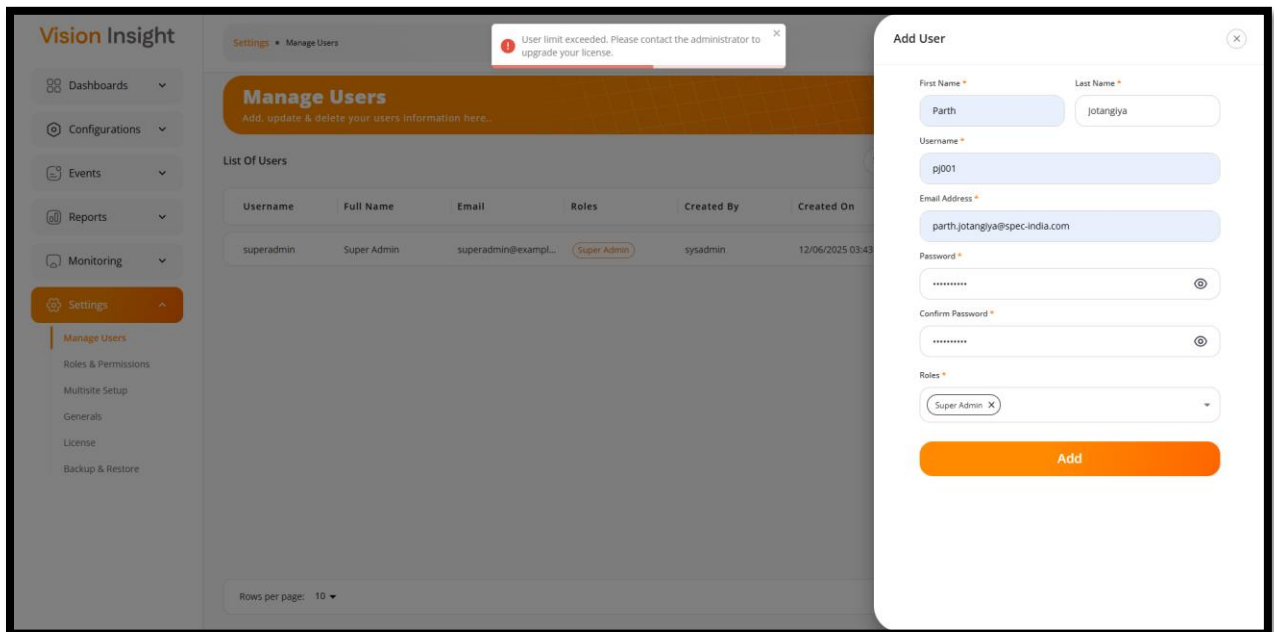


➔ The system will display an **appropriate validation message** when a user attempts to **enter a duplicate email address or username** that already exists in the application. This ensures that there are **no duplicate entries** with the same **email address or username**, as these are used as **login credentials**:

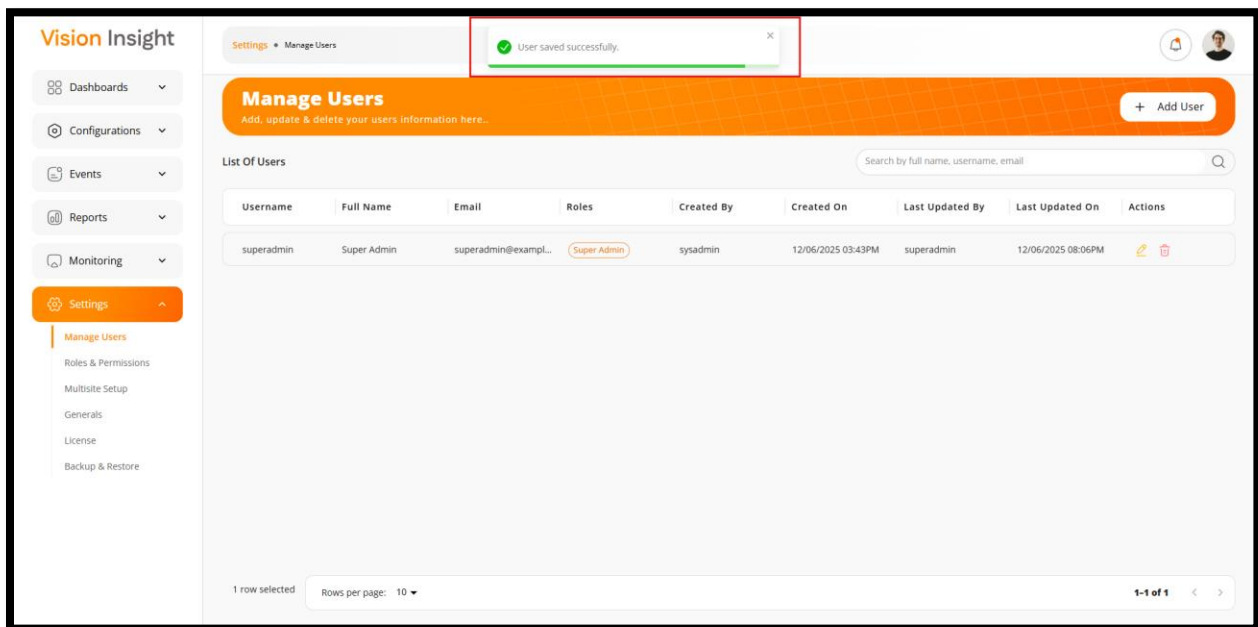
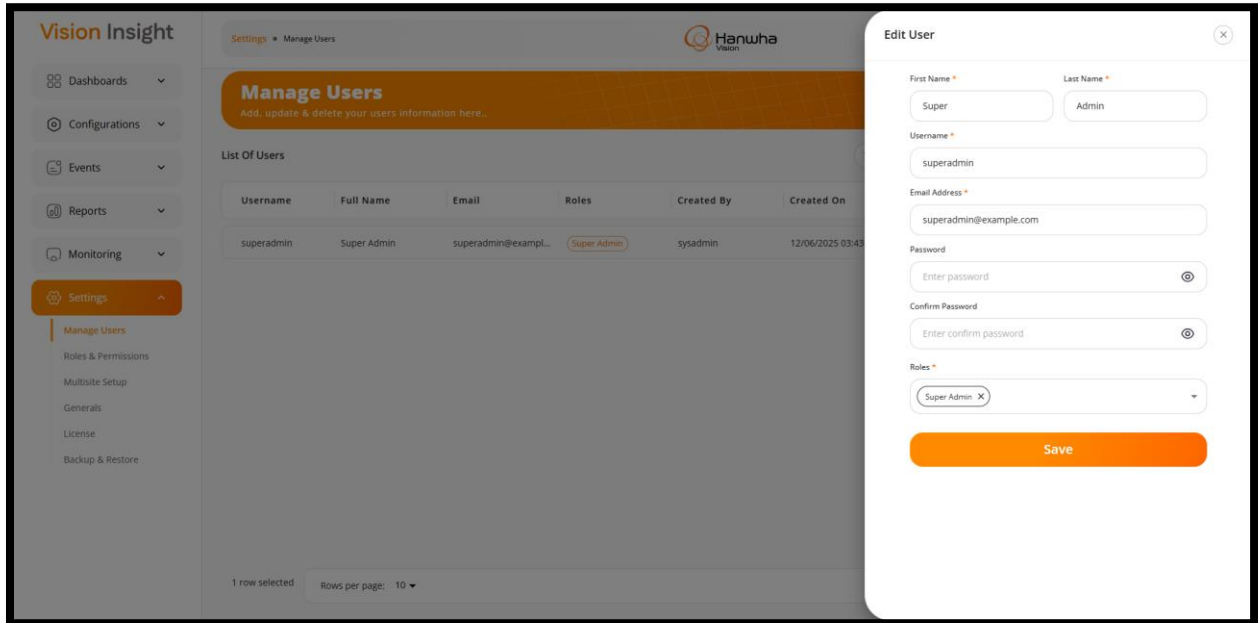




→ The system will display the **following message** when a user attempts to **add a new user**, but the **license limit for the number of allowed users** has been exceeded:



→ **Authorized users** can **edit any user's information** using the form shown below. Upon **successful update**, the system will display the **following success message**:

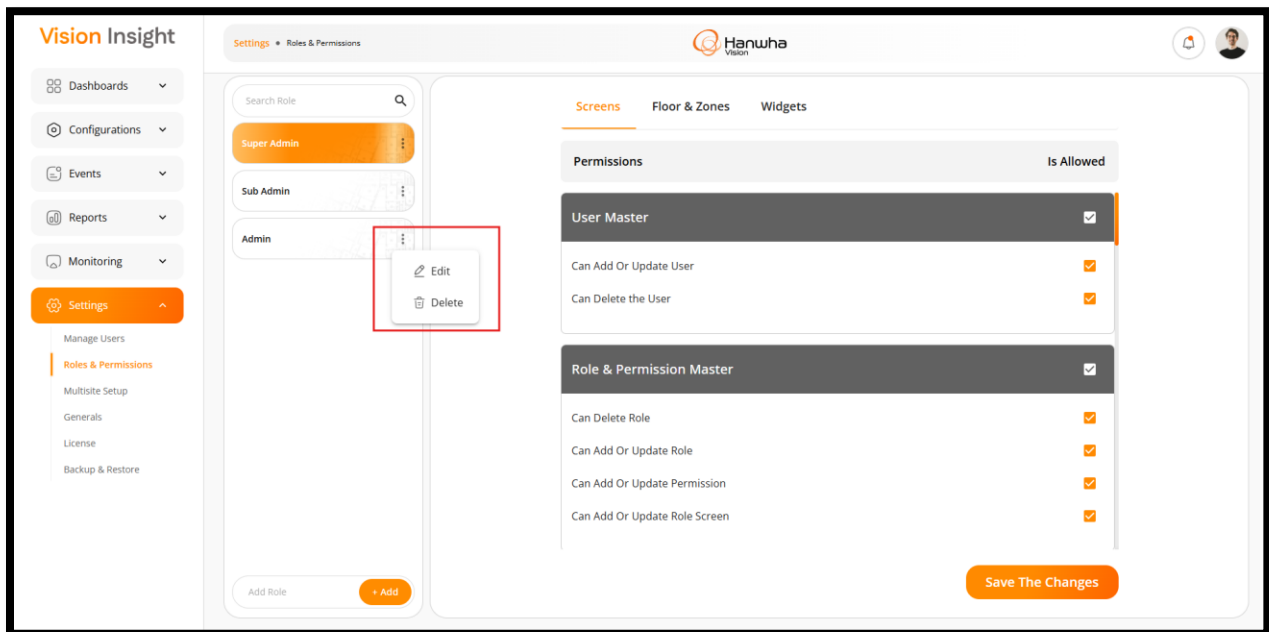
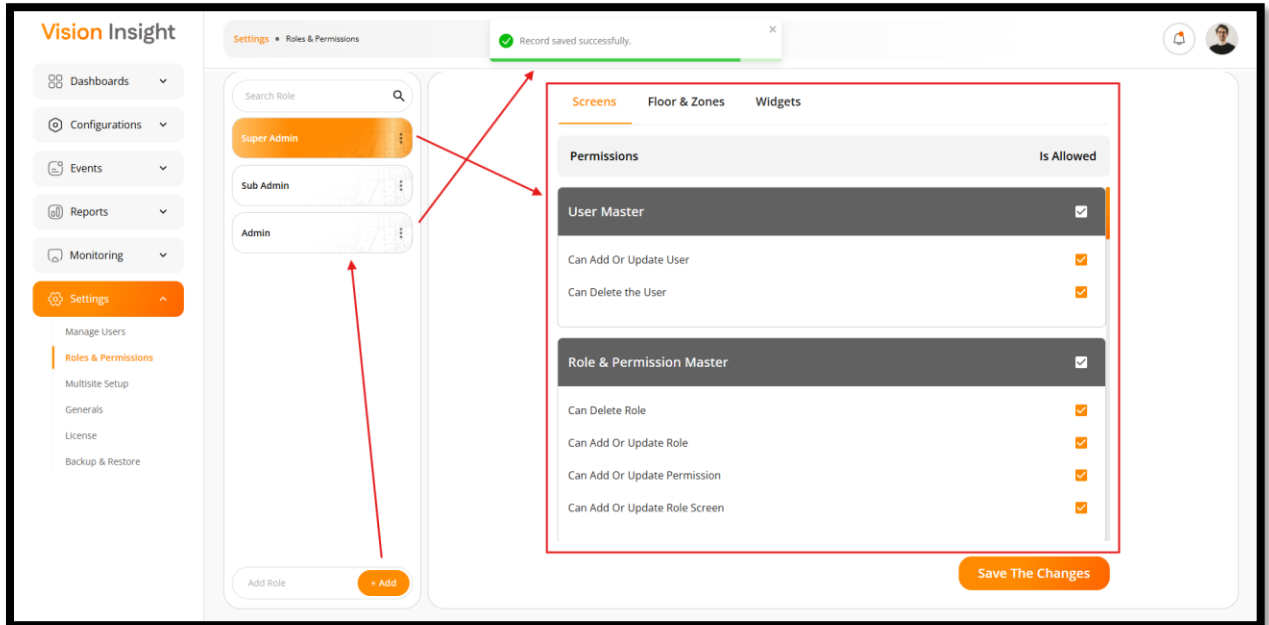


3) Role-based access control with permission management

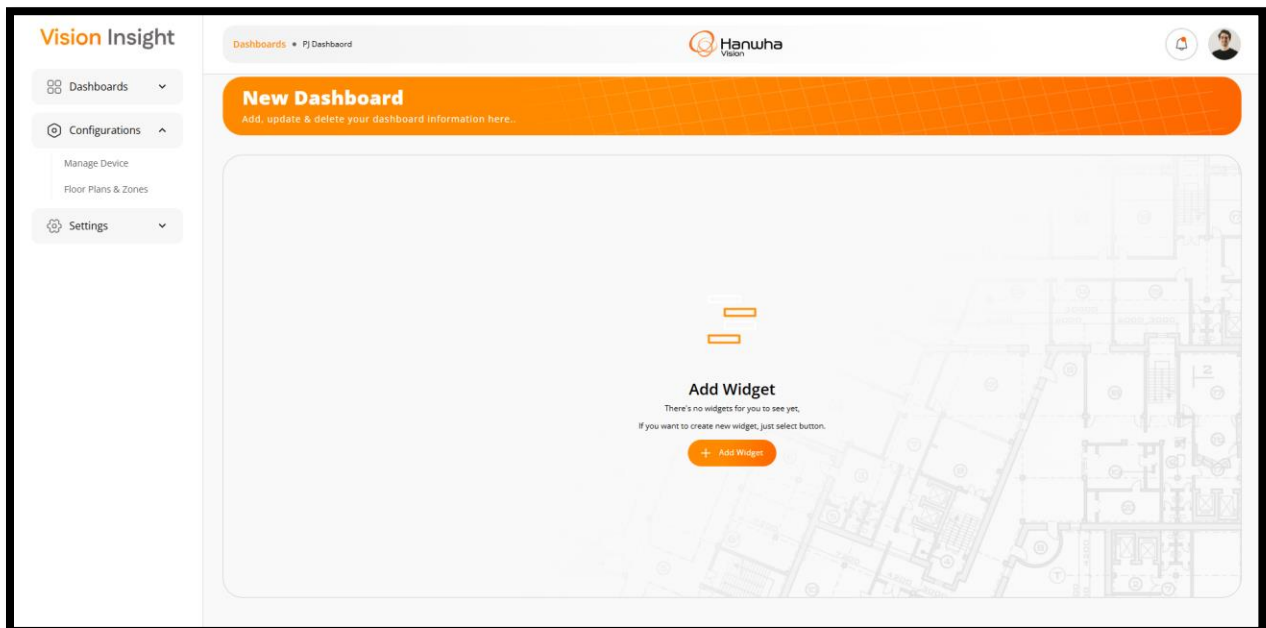
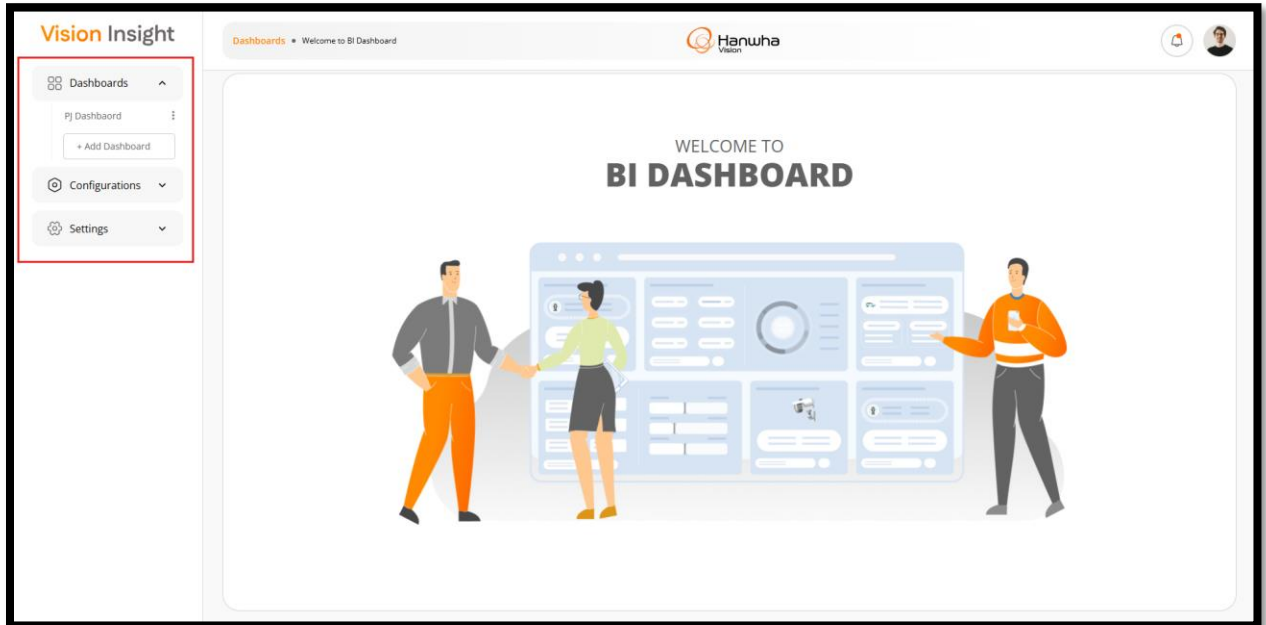
A system will be in place to define different roles and assign specific permissions to each role. This allows for granular control over what actions users can perform based on their assigned role(s).

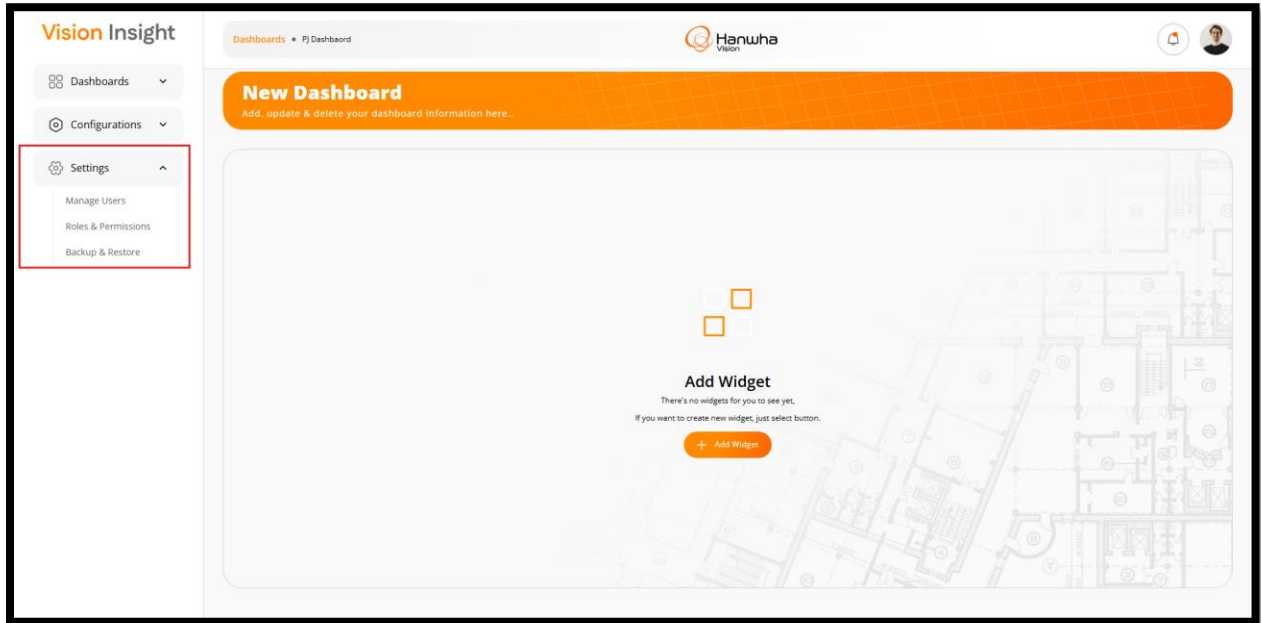
Screenshots:

- ➔ The screen below displays the **list of all created roles**, along with options to **Search**, **Add**, **Edit**, and **Delete** roles. The system will display an **appropriate success message** when a **new role is successfully created**, or **existing role is successfully updated**.
- ➔ Additionally, users can **assign permissions** at the **Screens**, **Floor & Zones**, and **Widgets** levels.



- ➔ When a user **logs into the system**, they will be able to **access only those pages and functionalities** that are permitted based on their **assigned role permissions**.
- ➔ In the image below, permissions have been assigned to the **"Admin" role** to access the following pages:
 - Dashboards
 - Manage Devices
 - Floor Plans & Zones
 - Manage Users
 - Roles & Permissions
 - Backup & Restore



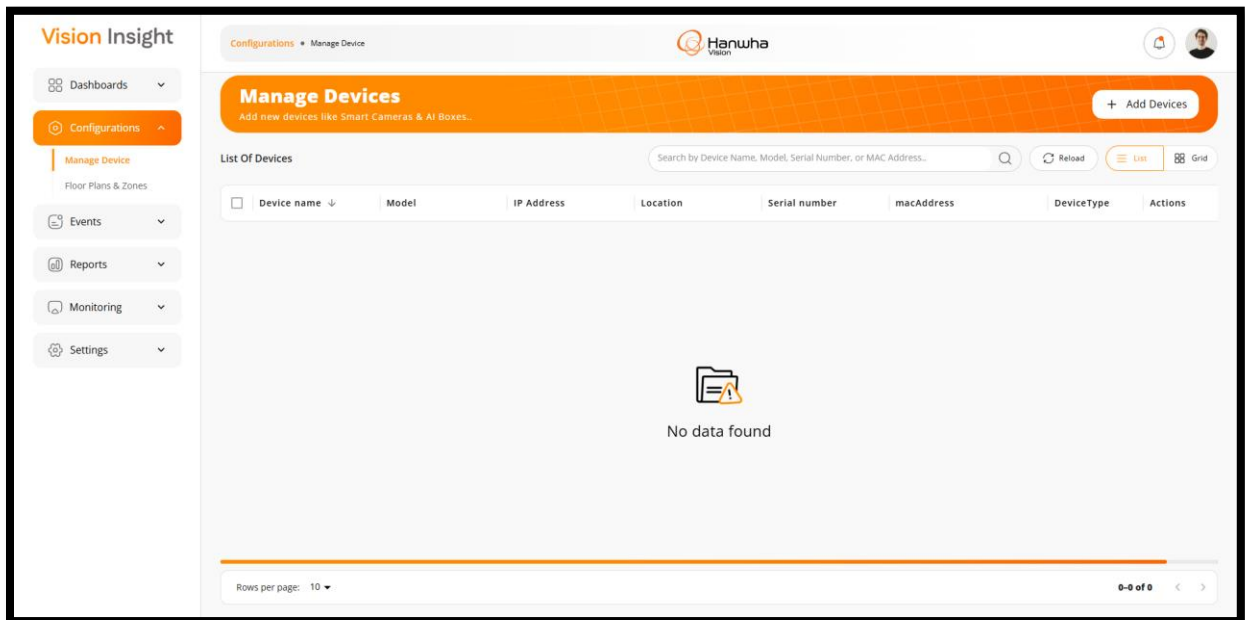


4) Functionality to manage camera devices and AI boxes

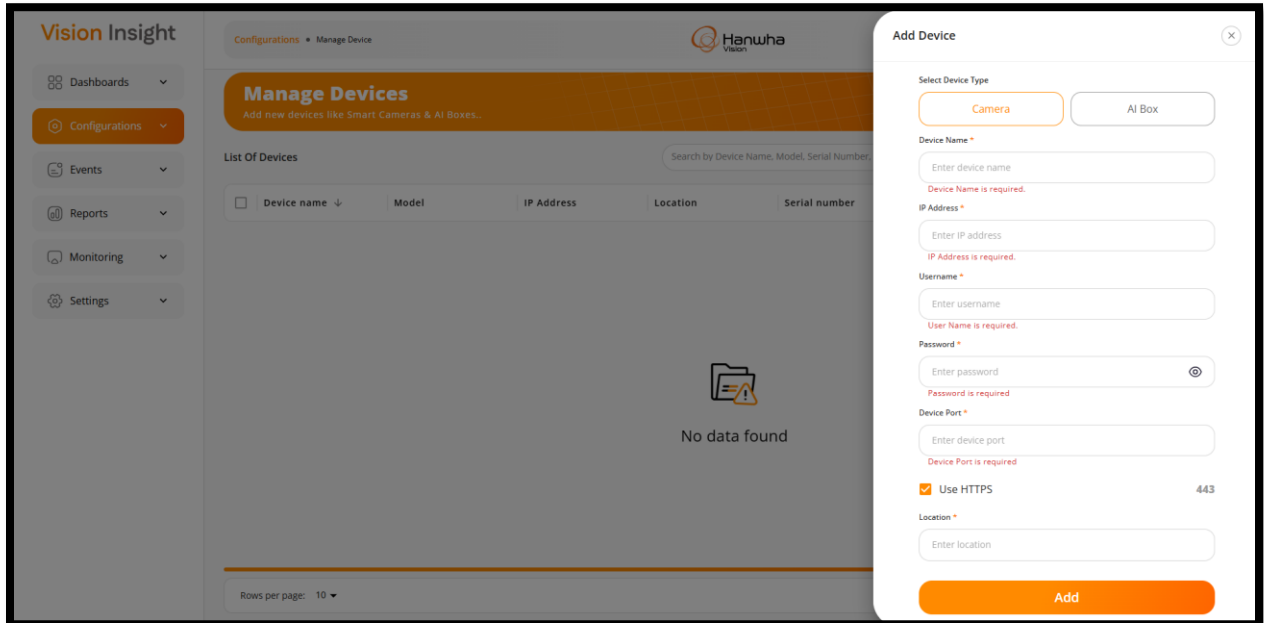
Authorized users will have the capability to create, modify, and delete devices like camera & AI Box.

Screenshots:

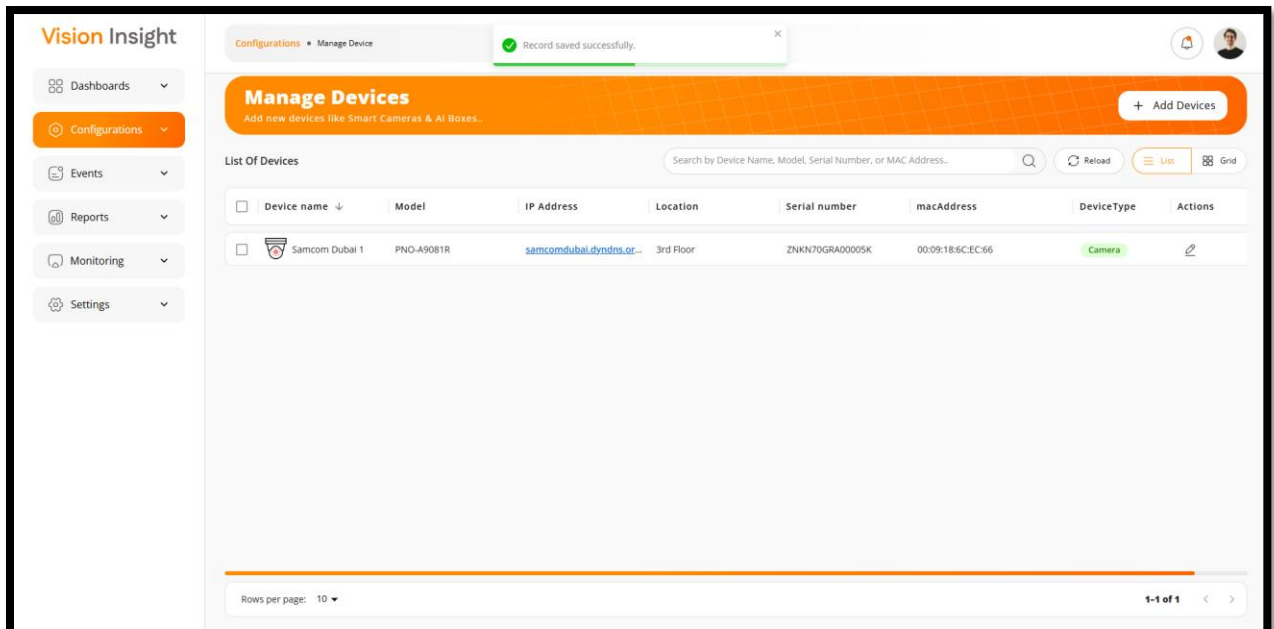
➔ The below screen will be displayed when **no device** is found in the **system**.



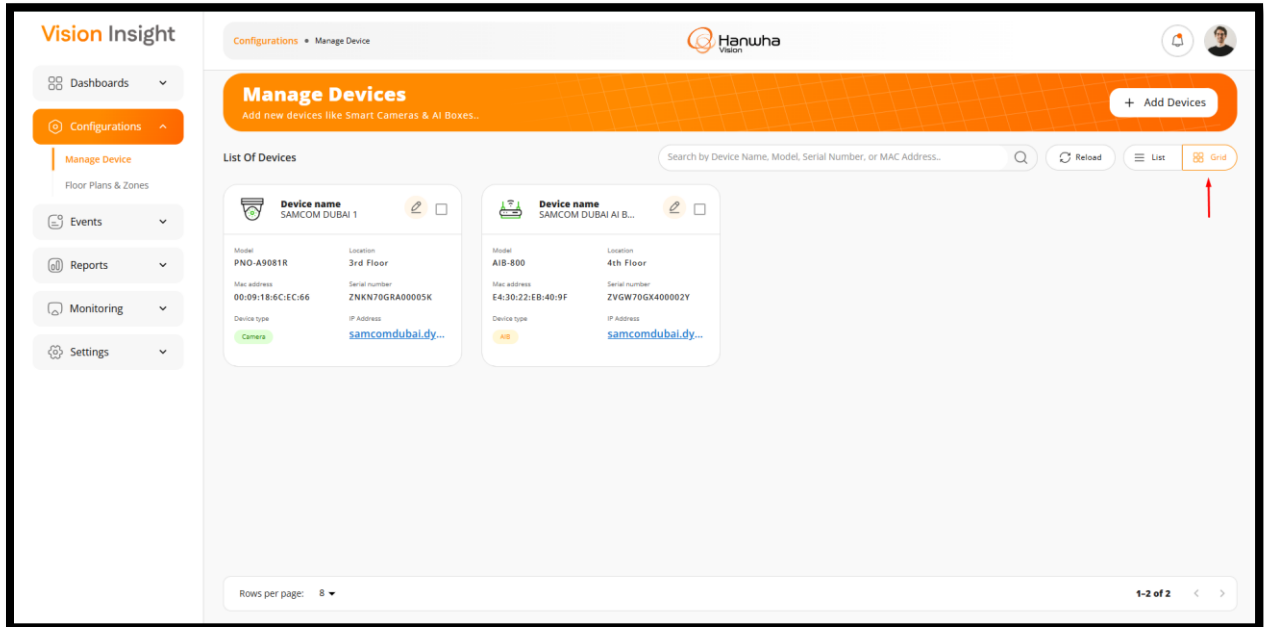
➔ The system will perform **all necessary validations** when **adding a new device** to ensure the entered information is valid and accurate.



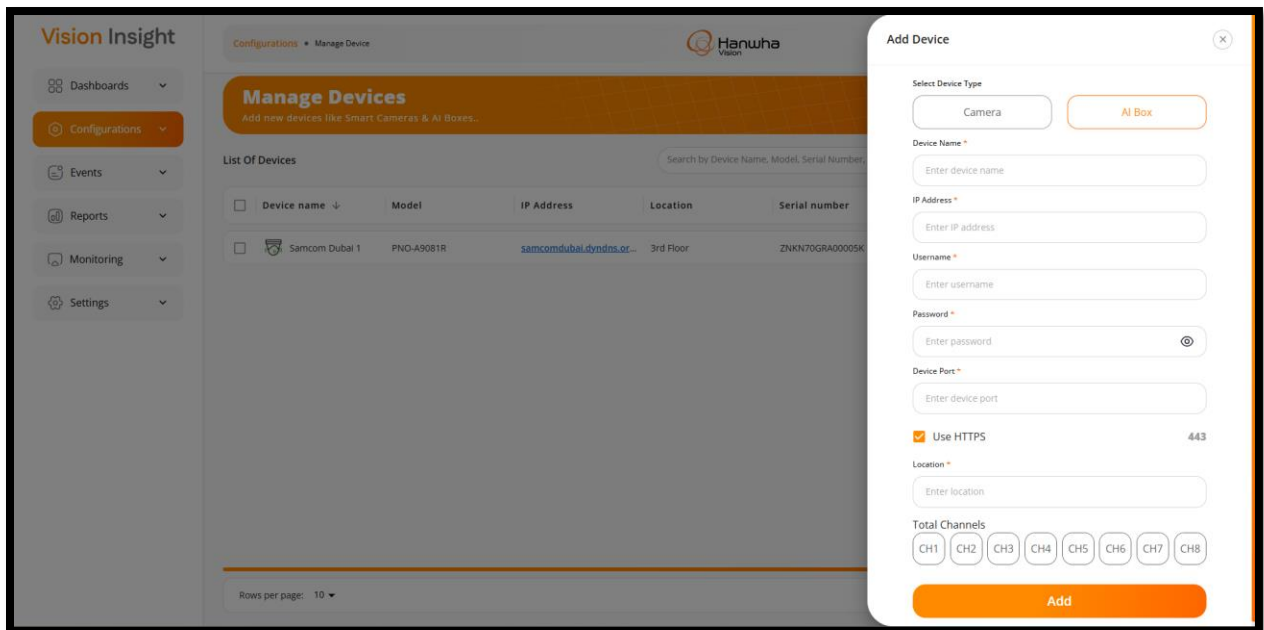
- ➔ The system will display the **following success message** when the **device is successfully added** to the system.
- ➔ While **adding a new device** (Camera or AI Box), the system will **check the current availability** of the device. If the device is **available/online**, it will be **added directly** to the system. Otherwise, the system will display the following message to the user:
“This device is currently unavailable or offline. Please verify the connection and try again”



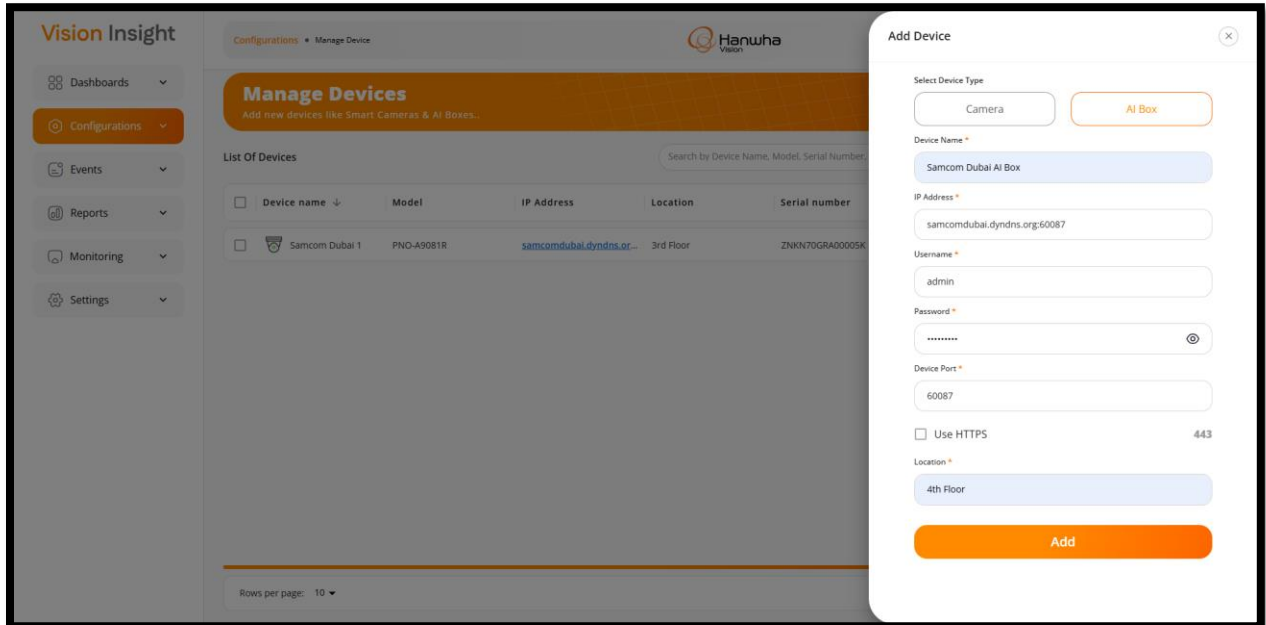
- ➔ The image below illustrates the **grid view** displaying **all added devices** in the system.



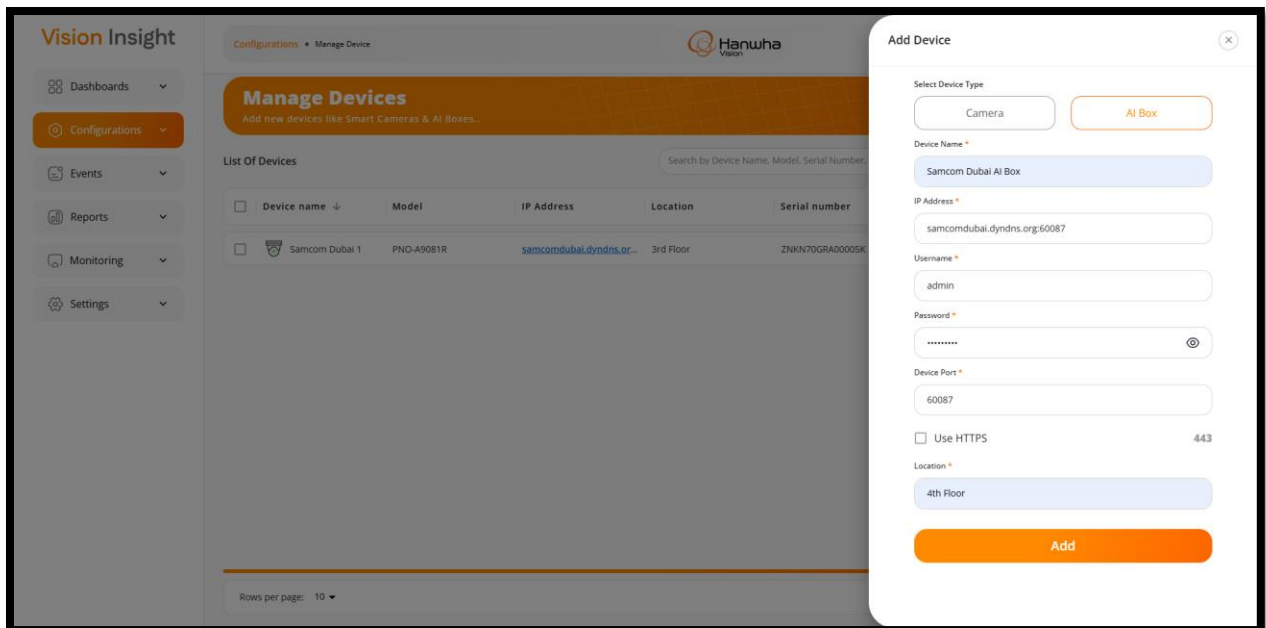
➔ The following fields will be displayed while adding an AI Box device to the system.



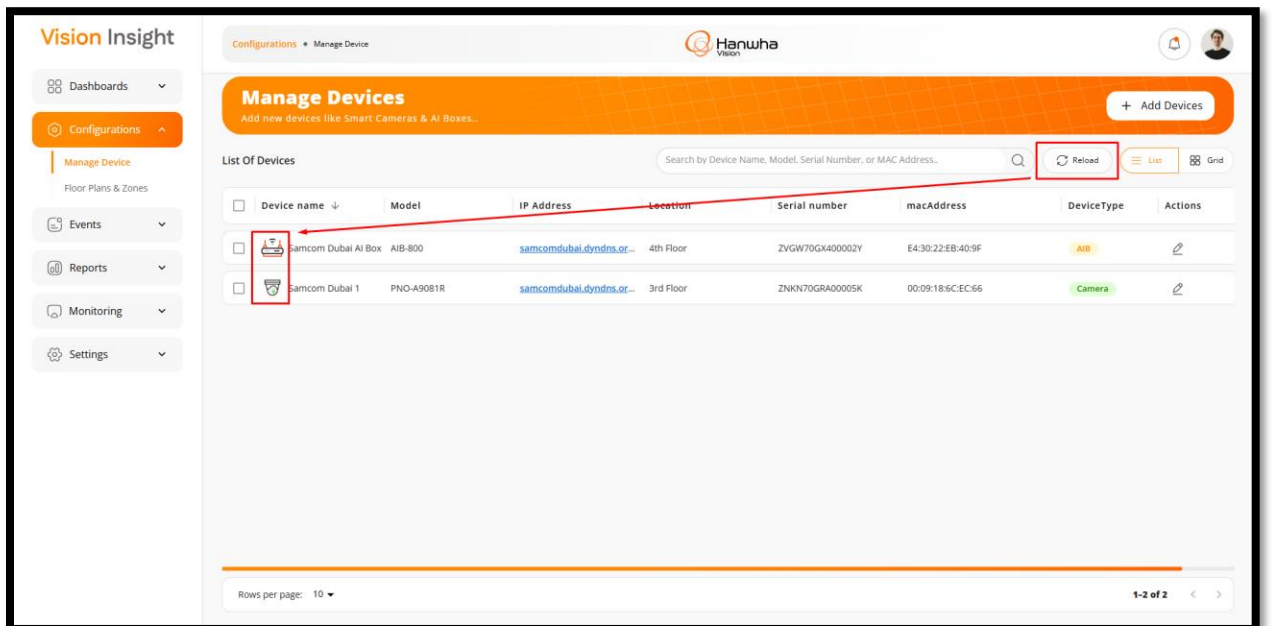
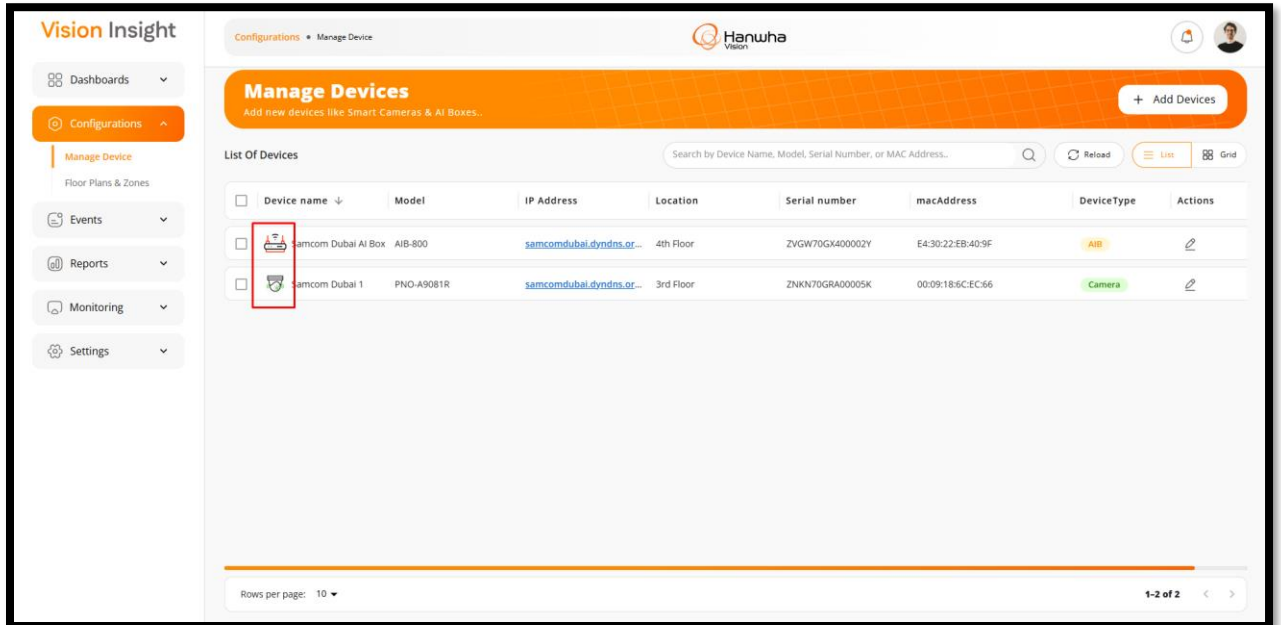
➔ The following fields will be displayed while adding an AI Box device to the system.

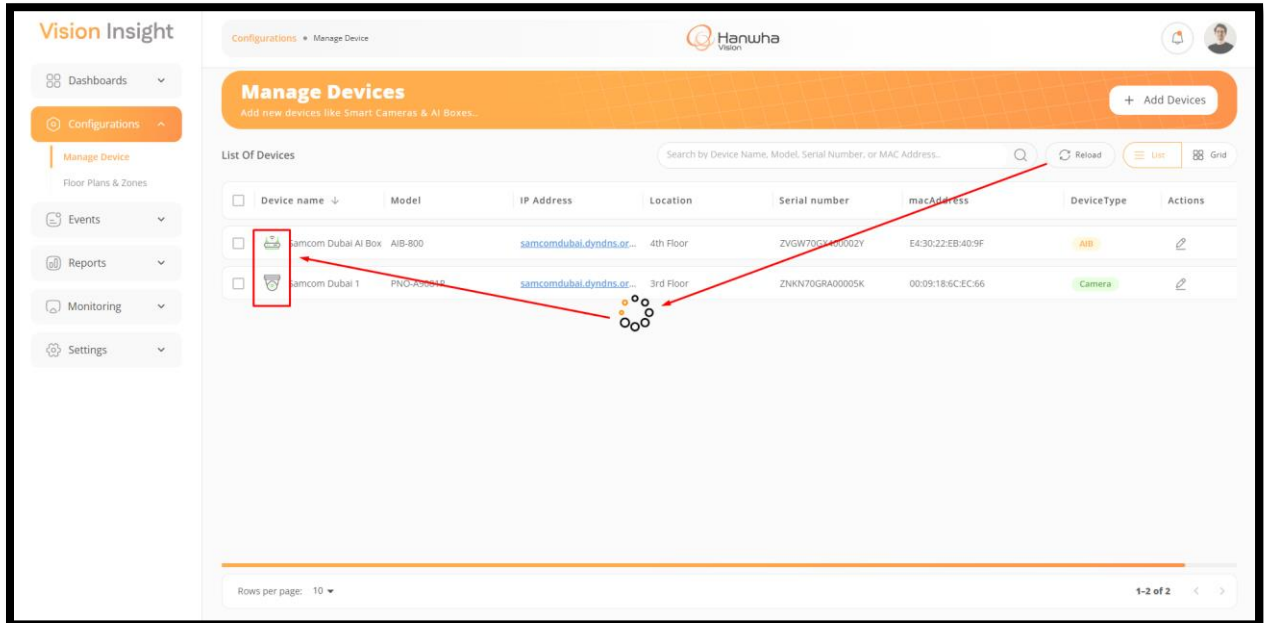


➔ The following fields will be displayed while adding an AI Box device to the system.

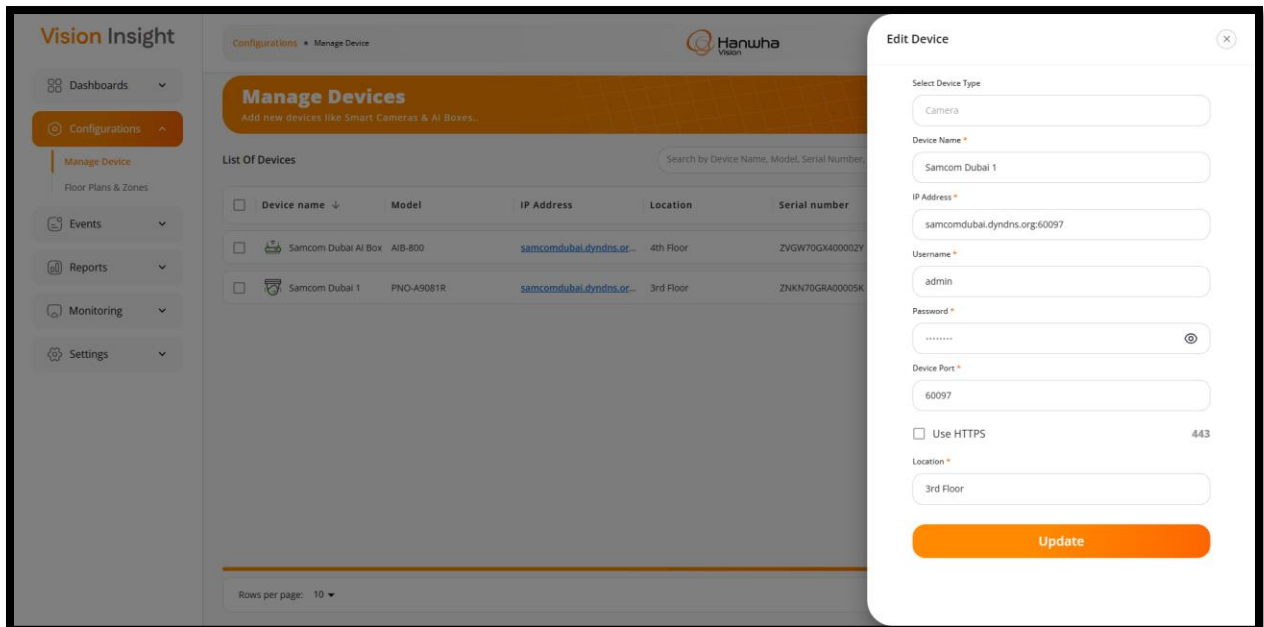


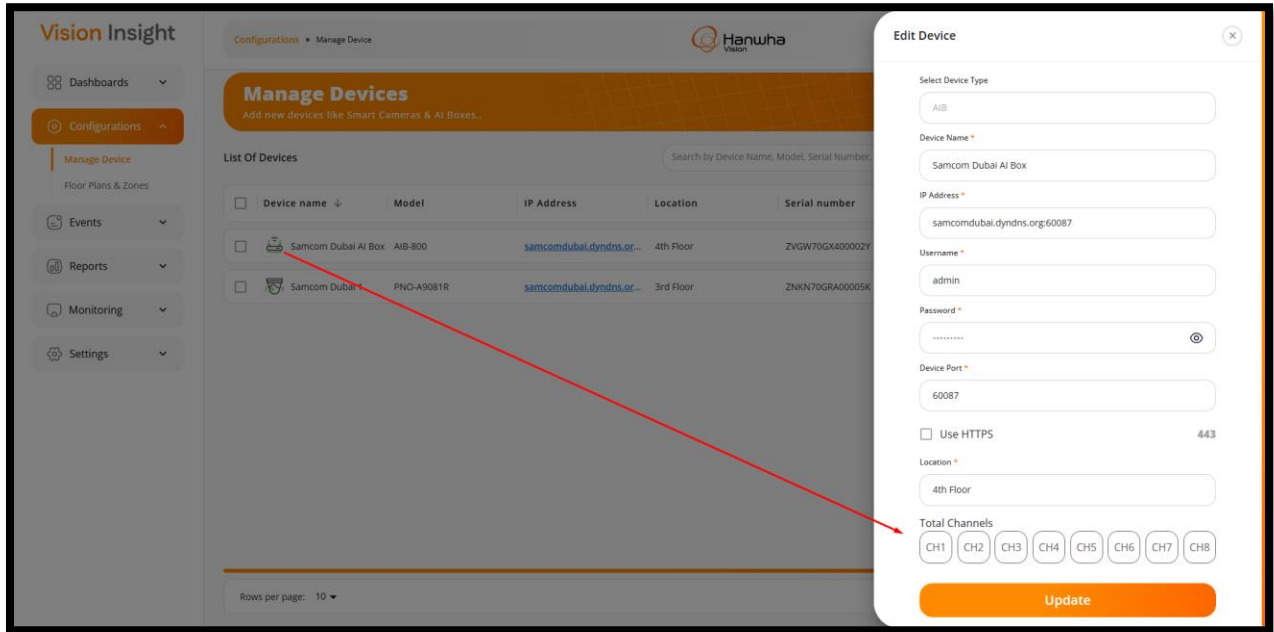
- ➔ The system will display **online and offline devices** as shown in the **highlighted section** of the screenshot below, so users can **easily identify** the device status directly from the **Manage Devices** page.
- ➔ If a device is **online**, it will be indicated in **green color**; if **offline**, it will be shown in **red color**.
- ➔ The system will make **real-time API calls** to check the **current device status**, and the updated status will be **reflected on the screen** accordingly when the user clicks the **"Reload"** button.



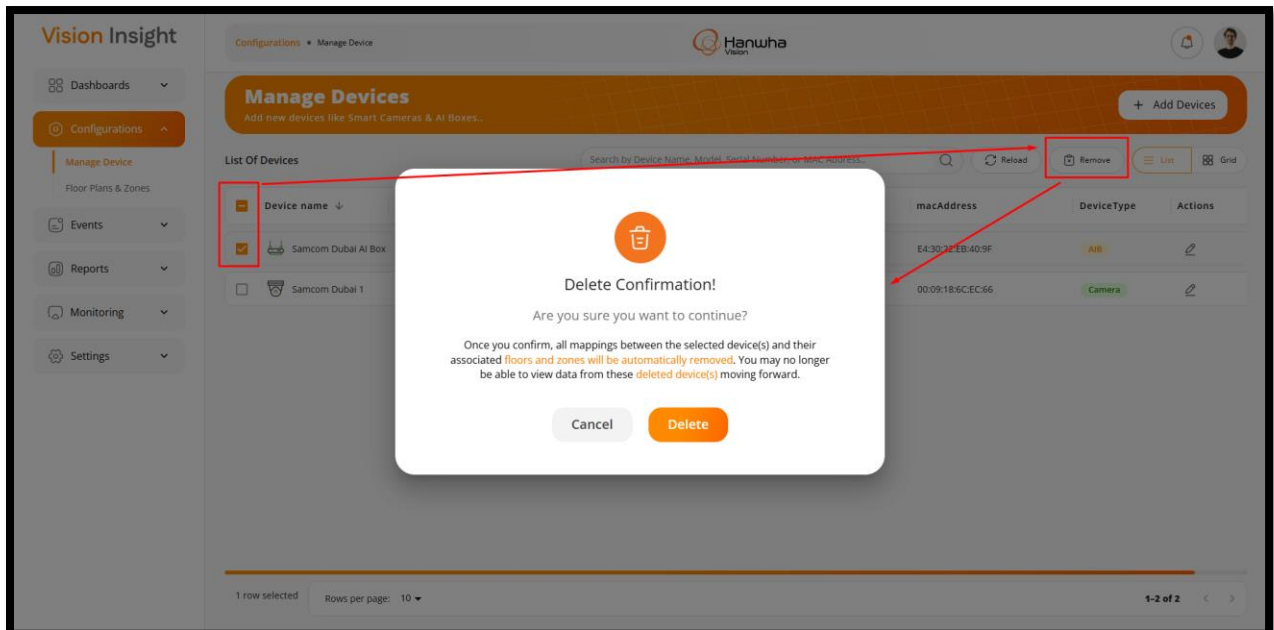


- ➔ Users can **edit device information** by clicking the **Edit** button from the list. The **details shown below** will be displayed for both **Camera** and **AI Box** devices.
- ➔ There is **only one difference** in the information displayed on the **Edit form** for Camera and AI Box devices:
 - The system will display the **"Total Channels"** field **only for the AI Box**, while this field will **not be shown** for Camera devices.





- ➔ The system will display the **“Remove”** button when the user **selects any record** by checking the **checkbox** from the list.
- ➔ Upon clicking the **“Remove”** button, the system will display a **confirmation popup** with a **clear message** to inform the user before removing the device from the system.
- ➔ Upon clicking the **“Delete”** button, the system will **perform the action** and display an **appropriate success or failure message** to the user based on the outcome.

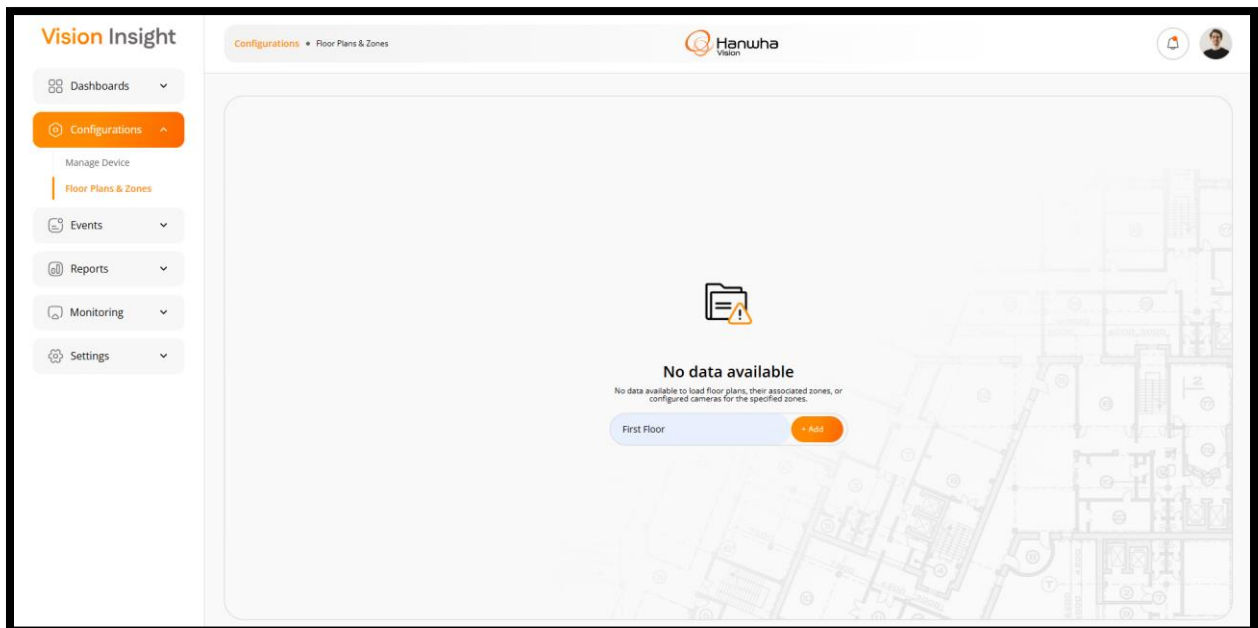


5) Floor plan and zone configuration capabilities & Mapping of cameras to designated zones

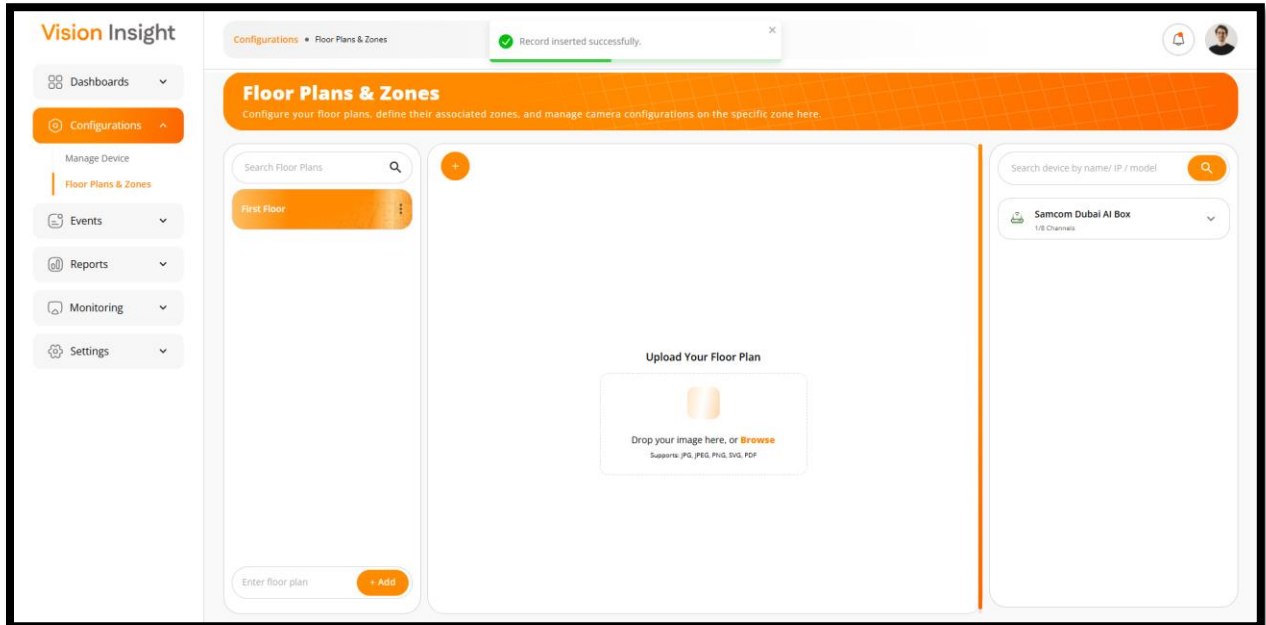
This feature includes the ability to **manage floor plans**, as well as to **define and configure different zones or areas** within those floor plans. Additionally, users can **assign and configure camera devices** to specific zones as needed.

Screenshots:

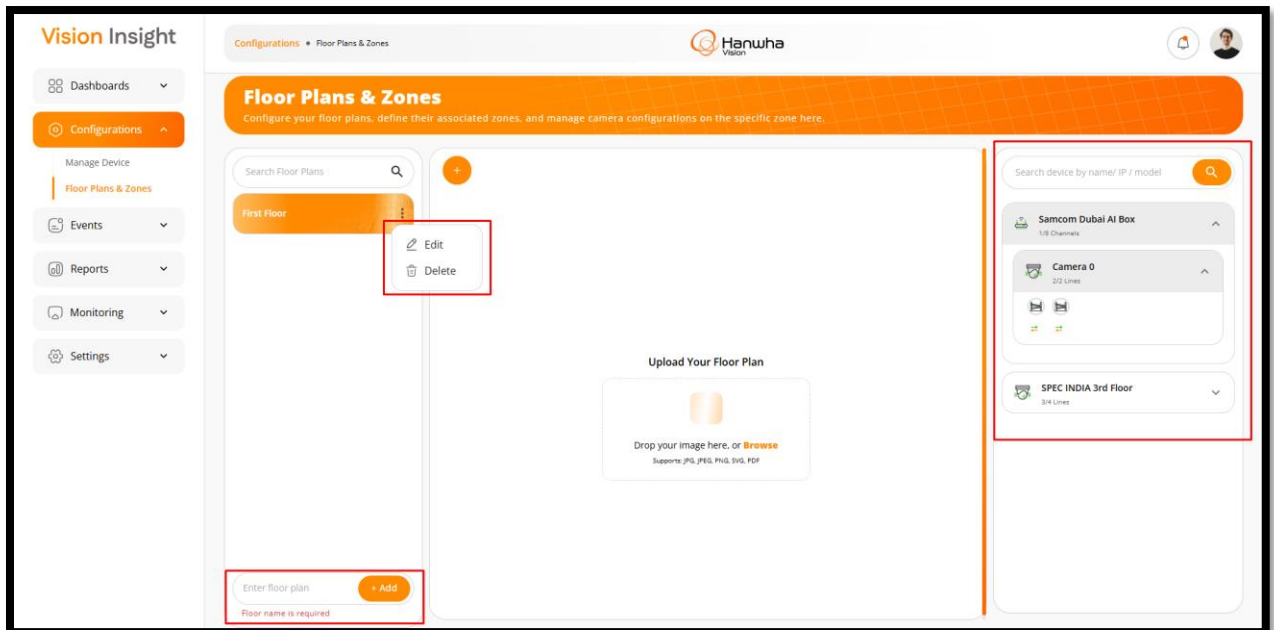
- ➔ The **screen below** will be displayed when **no floor plan has been created** in the system. Users can enter a **floor plan name** in the text box and click the “+ Add” button to **initiate the first floor plan** with any desired name.



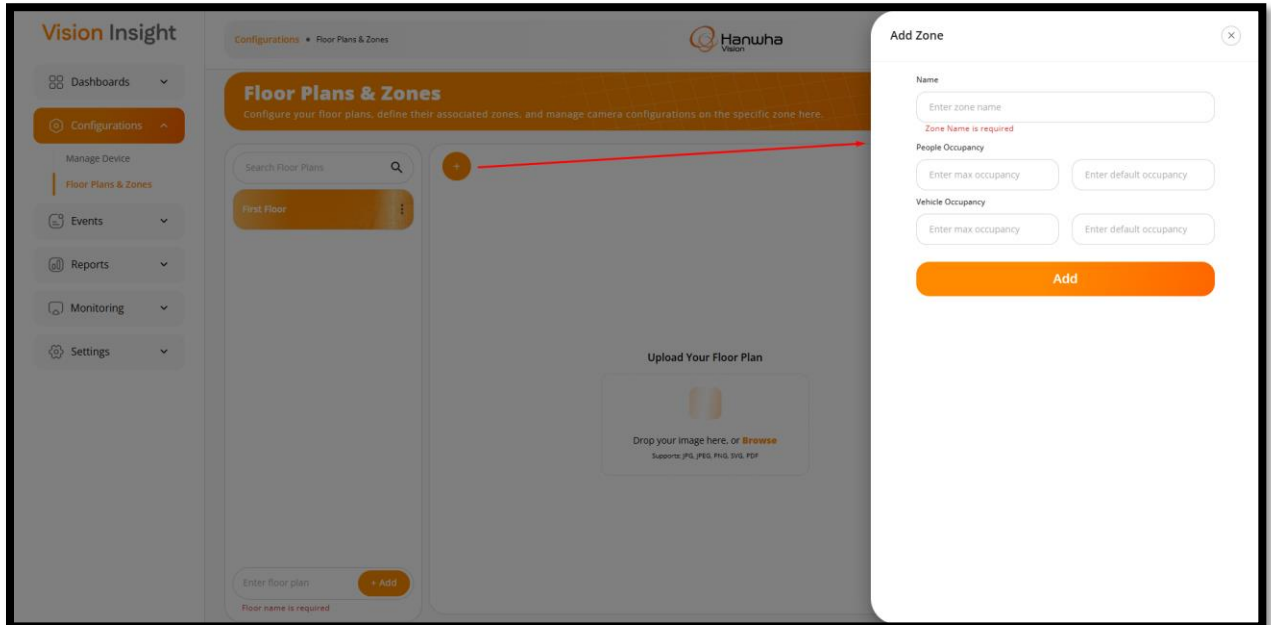
- ➔ The user will be **redirected to the screen below** after entering the floor plan name on the previous screen.
- ➔ The system will display an **appropriate success or failure message** based on the **response received** after submitting the request.



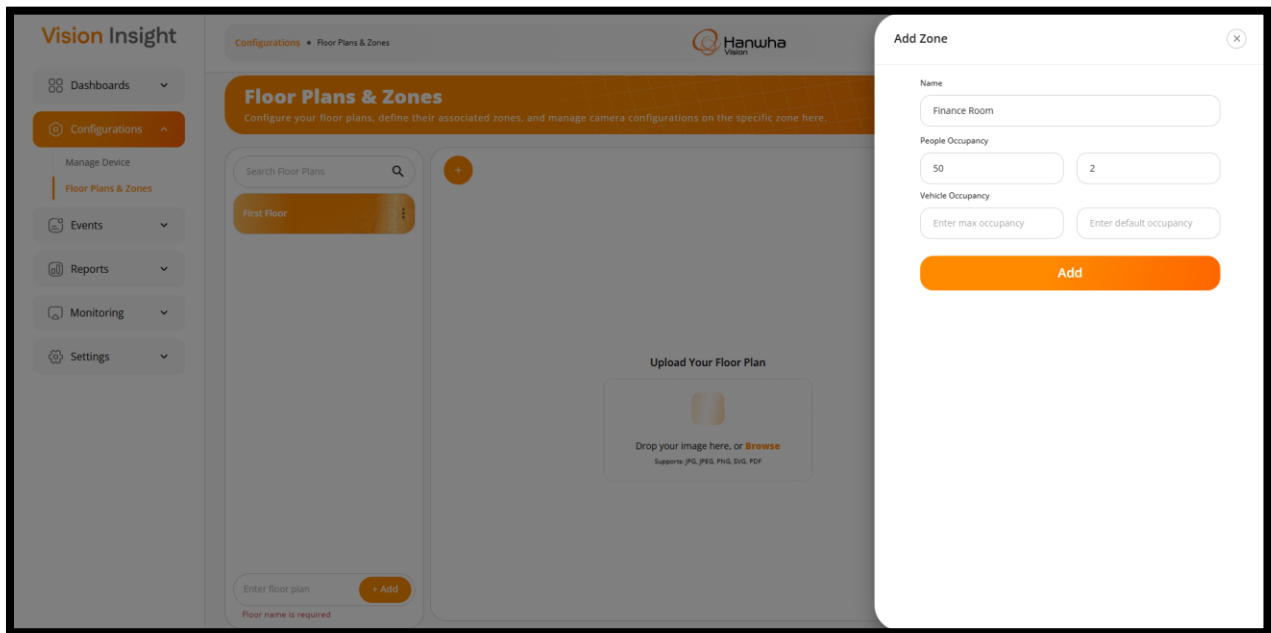
- ➔ The screen below displays a list of all created floor plans, along with options to **Search**, **Add**, **Edit**, and **Delete**.
- ➔ On the **right side**, the system shows all **online AI Boxes** with their **associated camera devices**, including those configured in the system from the “**Manage Devices**” screen.
- ➔ It also includes the option to **view associated active lines** of a specific camera device and **filter data** based on **Device Name**, **IP Address**, and **Model Name**

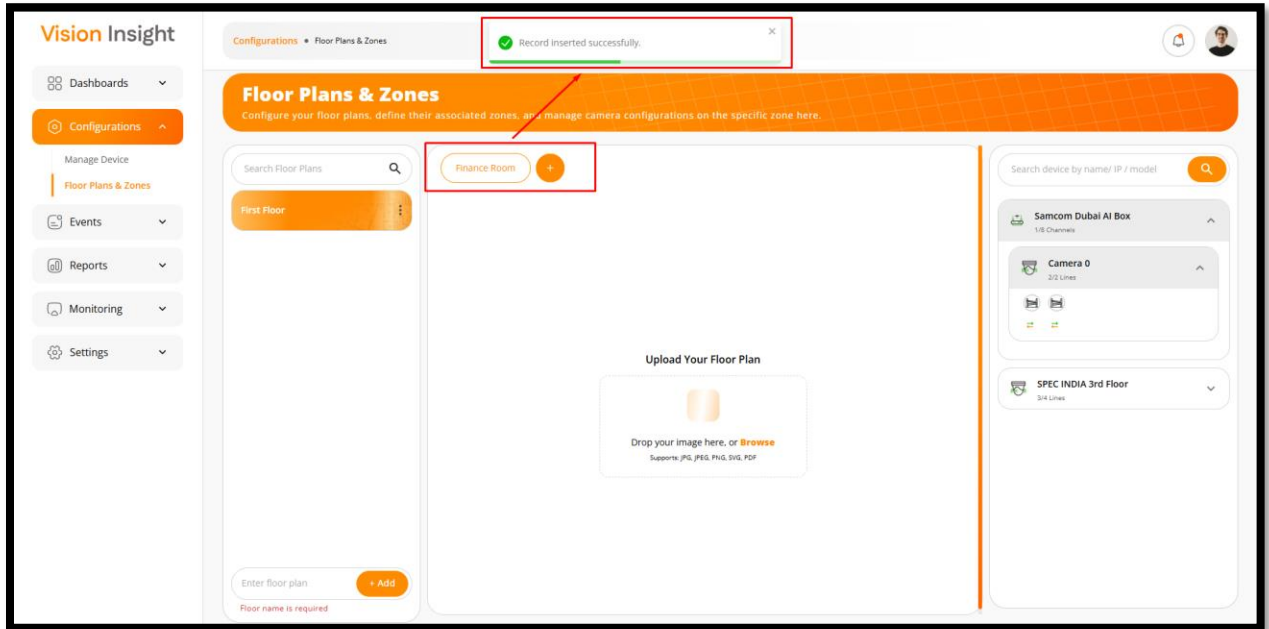


- ➔ The screen below displays the **form** used to **create a new zone** with the necessary fields on a specific **floor plan**.
- ➔ The system will **validate the entered information** and display **appropriate messages** when required fields are missing or incorrect.

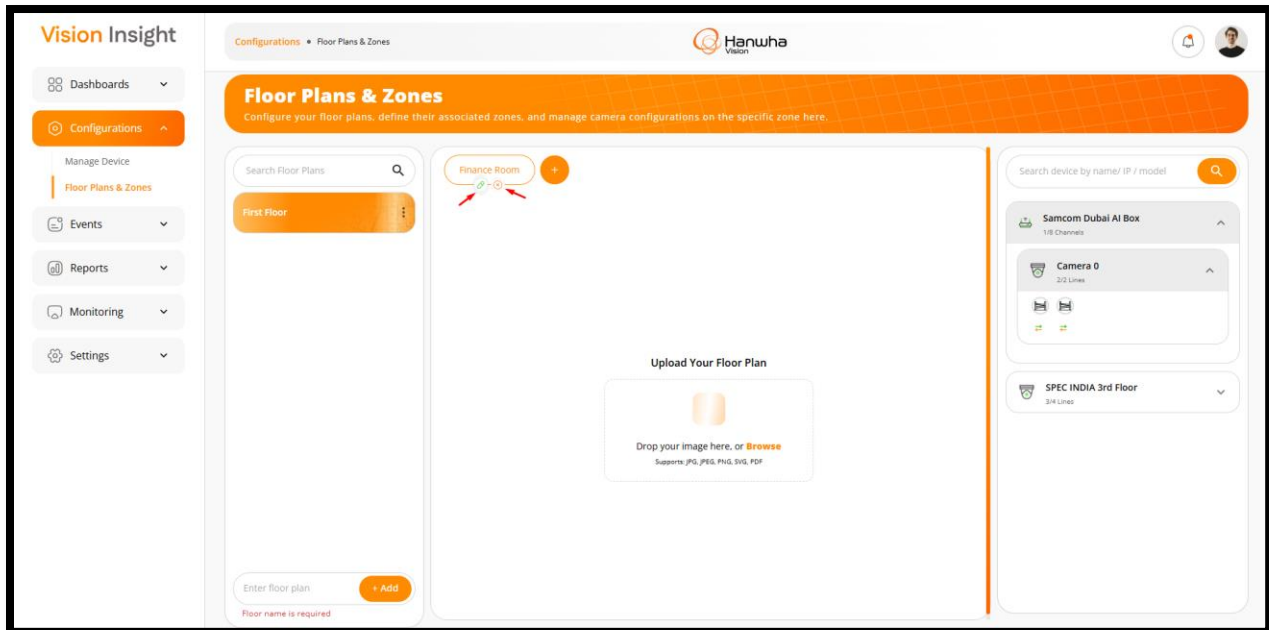


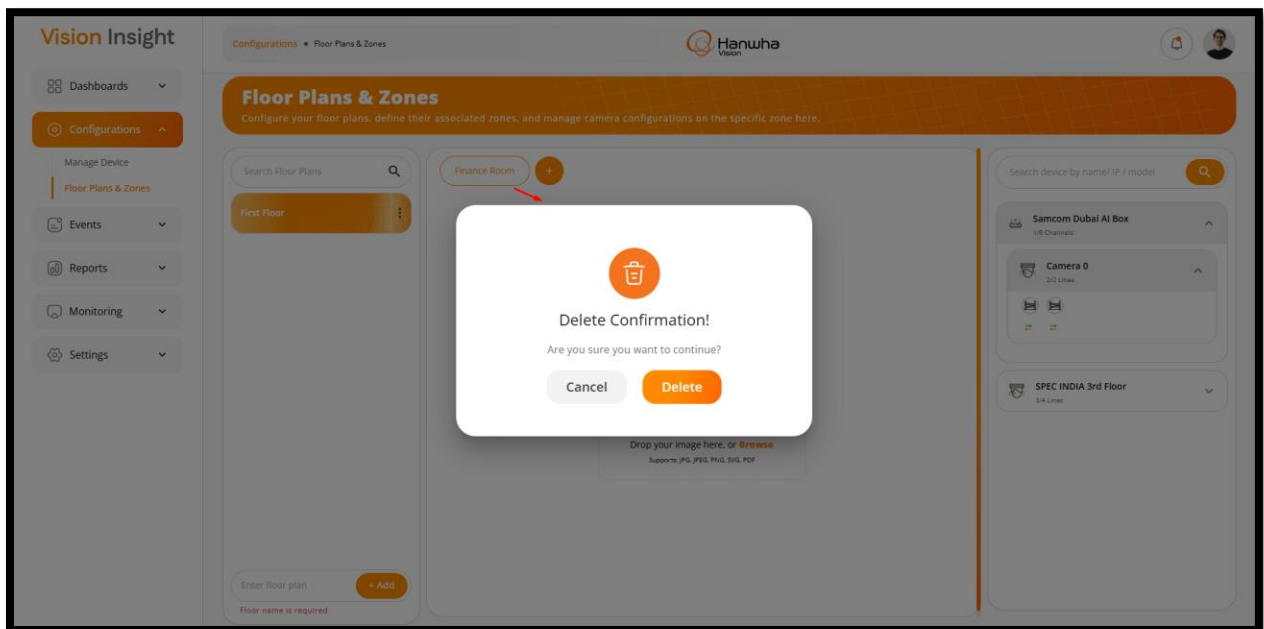
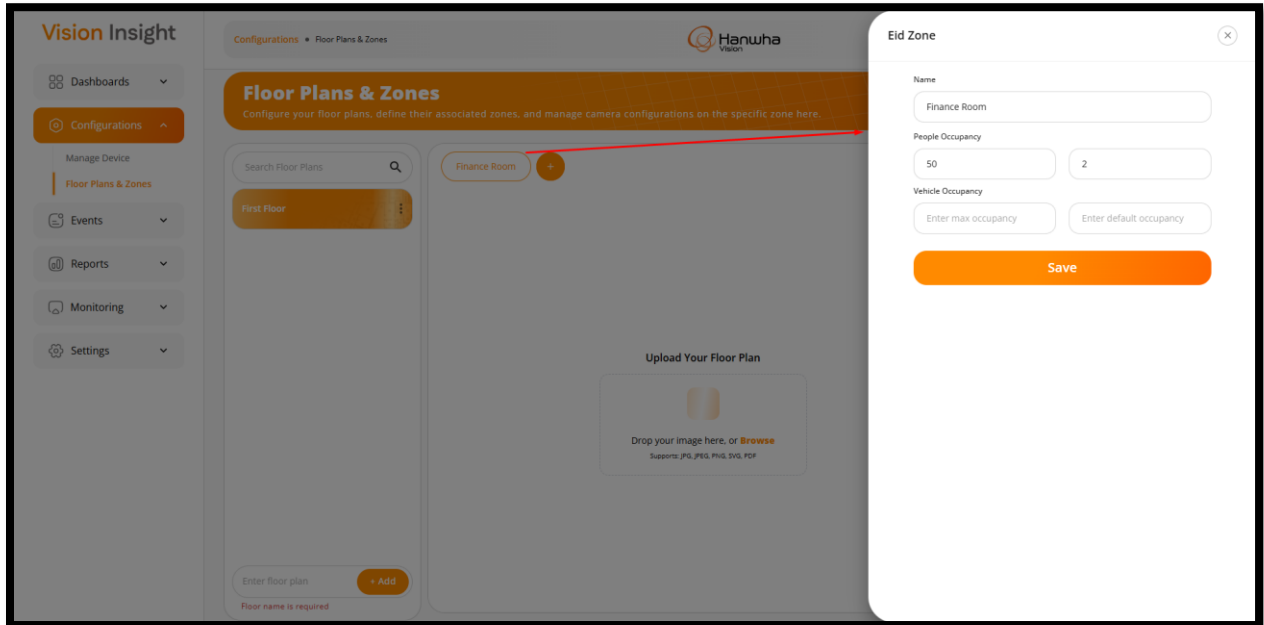
- ➔ Once the user fills in the necessary details in the “Add Zone” form, the system will **submit the information** and display an **appropriate success or failure message** based on the result. If the zone is **successfully added**, it will then be **displayed in the list** of existing zones.



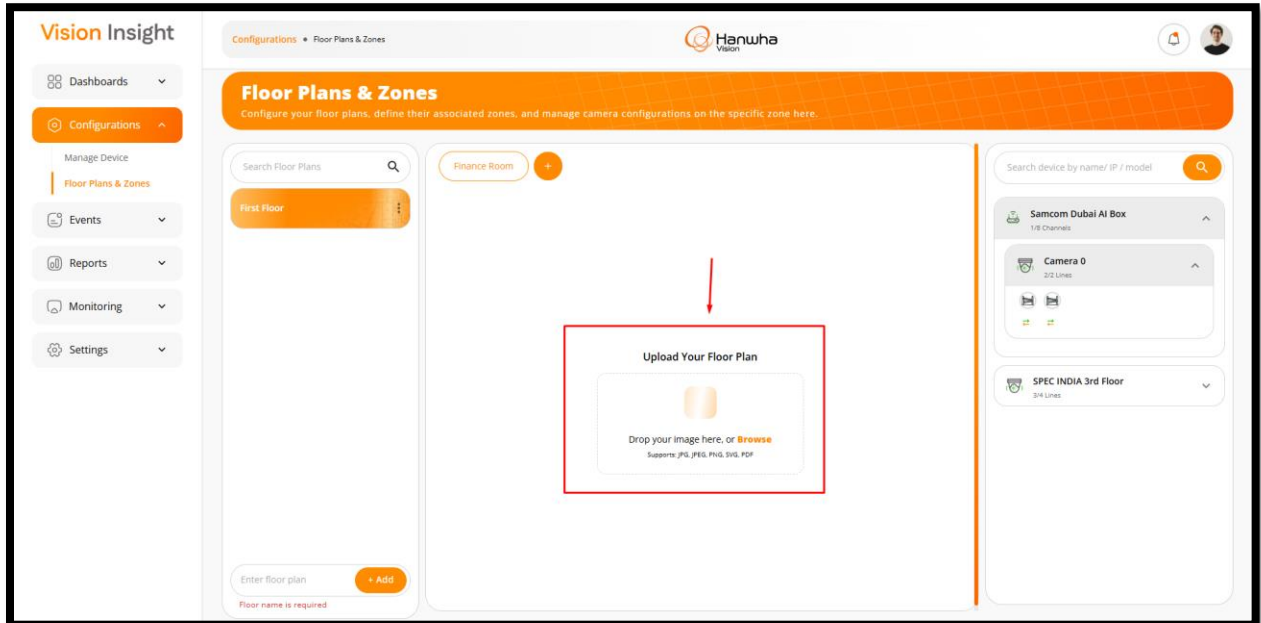


➔ Users can **edit or delete** zones by clicking on the **highlighted buttons** shown below.

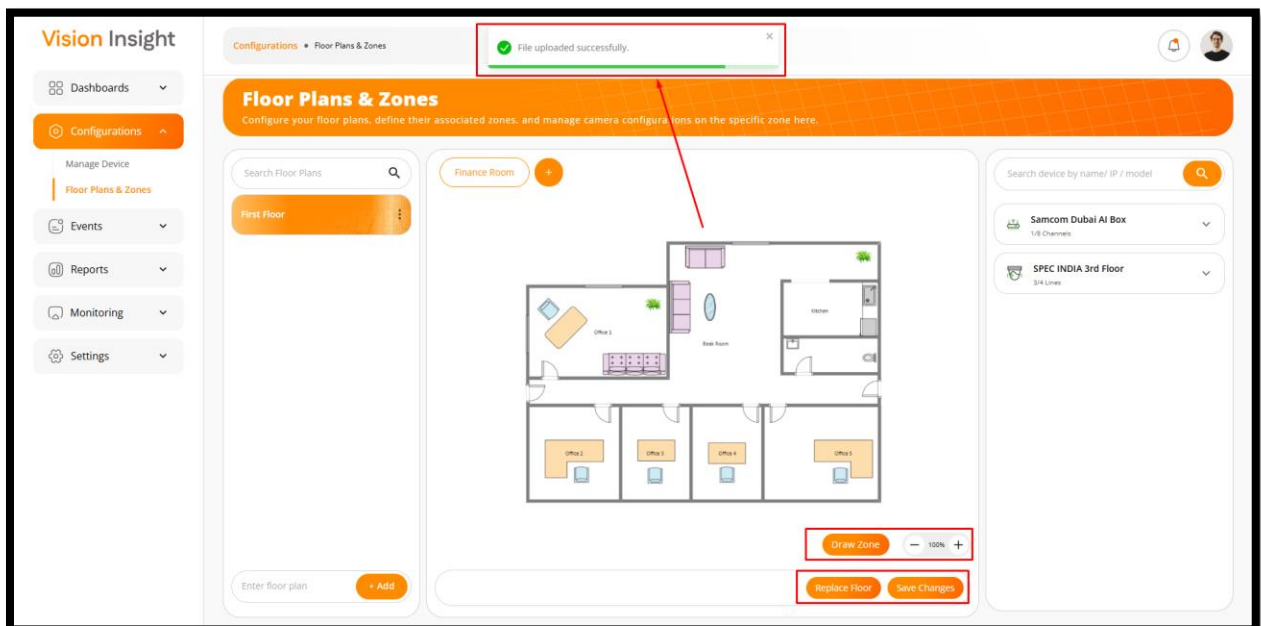


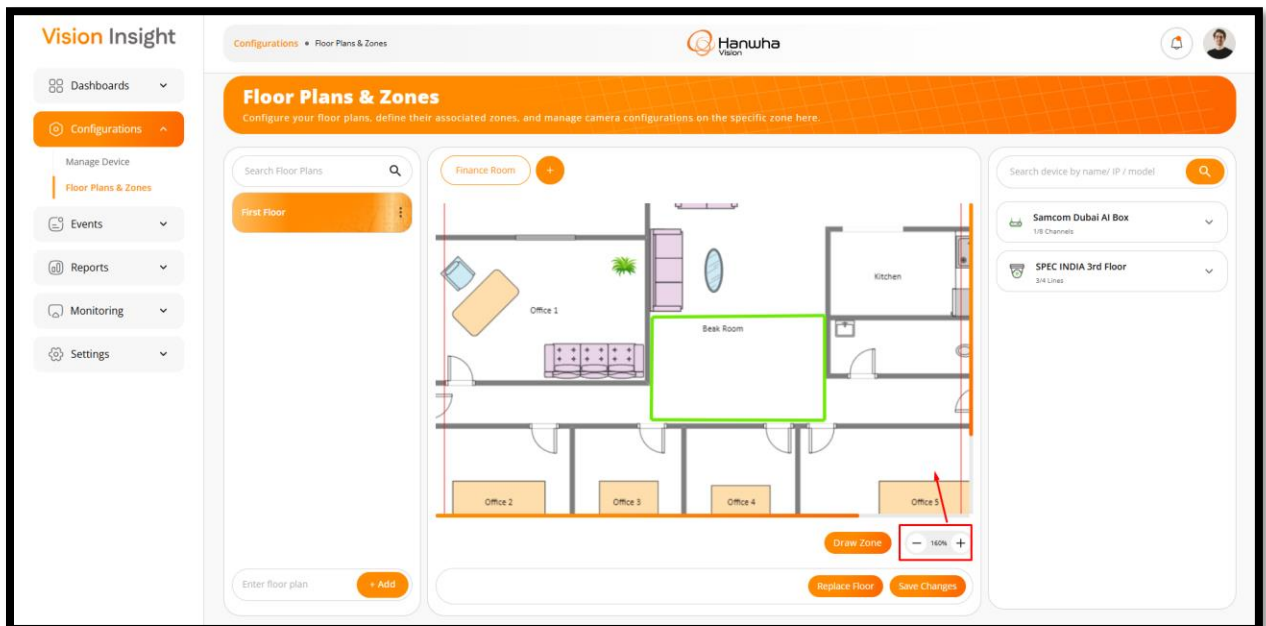
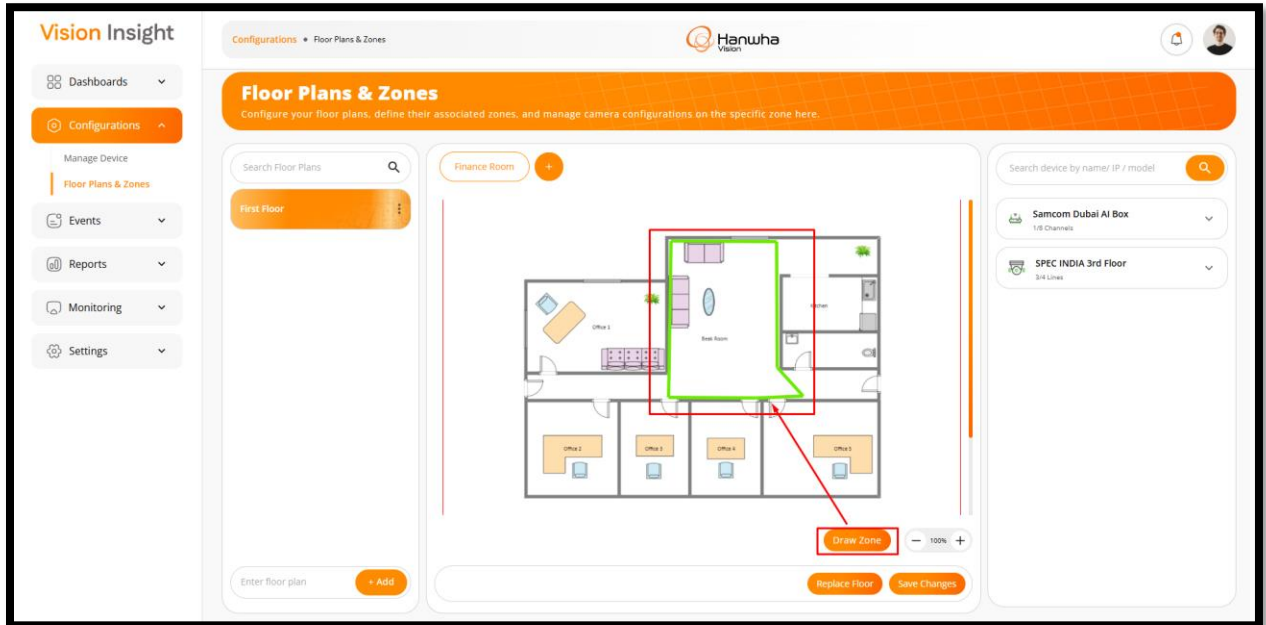


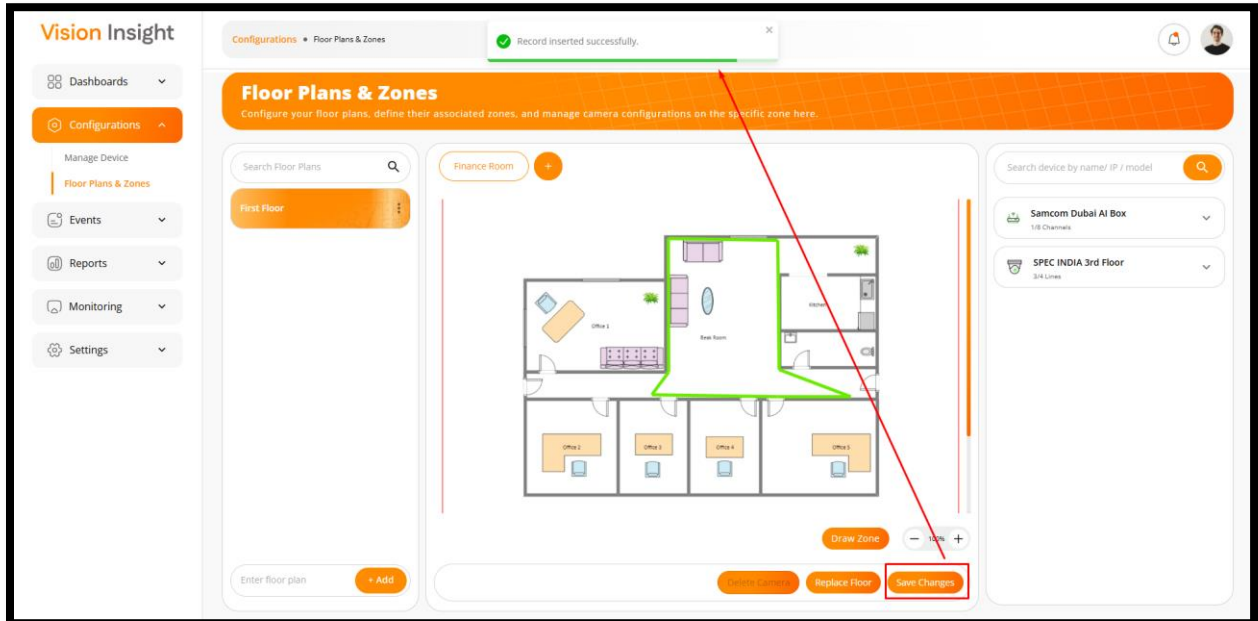
➔ Upon clicking the **highlighted area below**, users will be able to **upload a floor plan image** in the following supported formats: **PNG, JPG, JPEG, SVG, and PDF** only.



- ➔ The system will display an **appropriate success or failure message** upon uploading the **floor plan image**, based on the **response received**.
- ➔ After successfully uploading the image, the system will display the following **options**:
 - **Replace Floor**: Allows users to **replace the floor plan image** at any time.
 - **Save Changes**: Saves all **updated changes**. Users **must click this** after modifying any details on the floor plan image.
 - **Draw Zone**: Enables users to **draw zones** directly on the floor plan image.
 - **Zoom In & Out**: Allows users to **zoom in and out** of the floor plan image within the **specified area**.



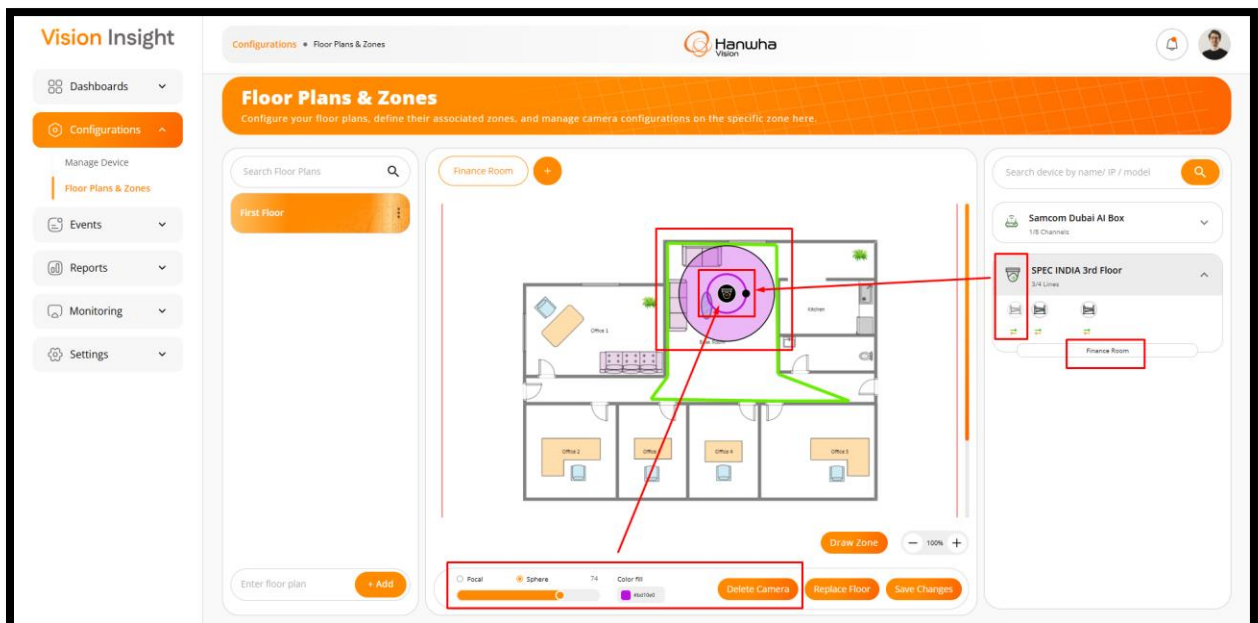




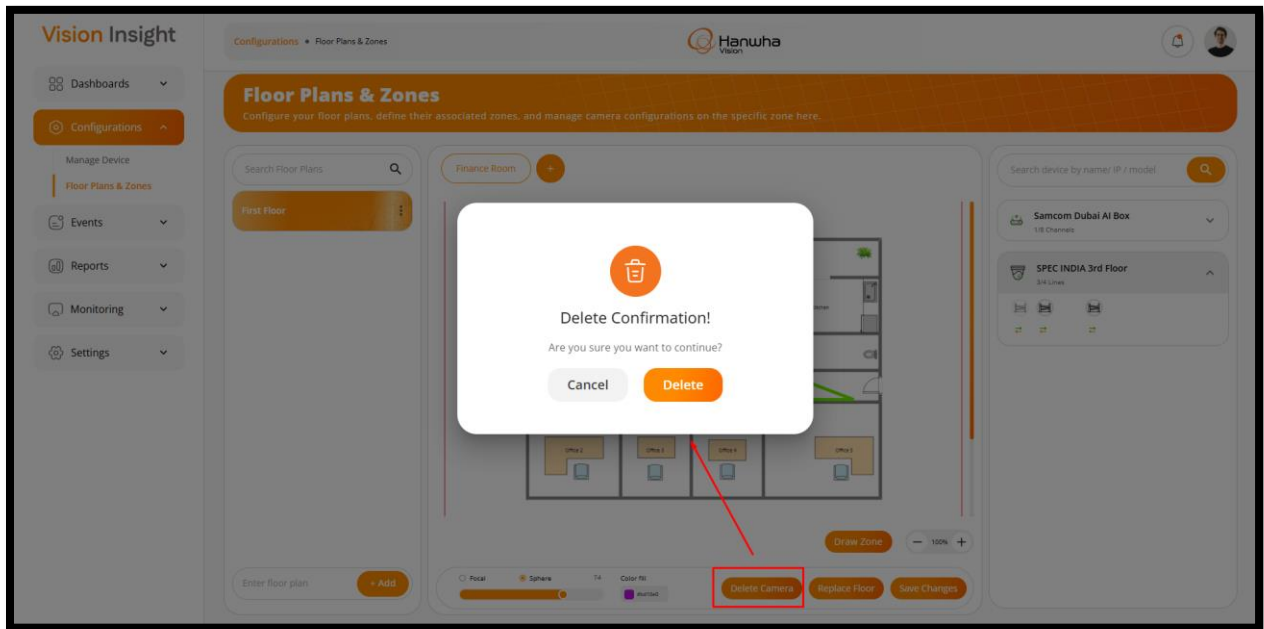
➔ The screen below illustrates the camera configuration options for a specific zone under a selected floor plan.

➔ Users can perform the following actions:

- **Drag & Drop** a camera from the **right-hand side** active camera or AI Box list, selecting the number of **line(s)** to be configured for the specific camera.
- The system will display the **associated floor plan and zone name** when the user **hovers** over a specific line of a camera in the list.
- **Rotate** the camera as needed.
- Select between **“Focal”** or **“Sphere”** modes.
- **Apply custom colors** to the selected **Focal** or **Sphere** area.
- Use the option to **delete** a camera from the configured plan.



➔ The system will display a **confirmation popup** when the user attempts to **delete a camera** by clicking the **“Delete Camera”** button.

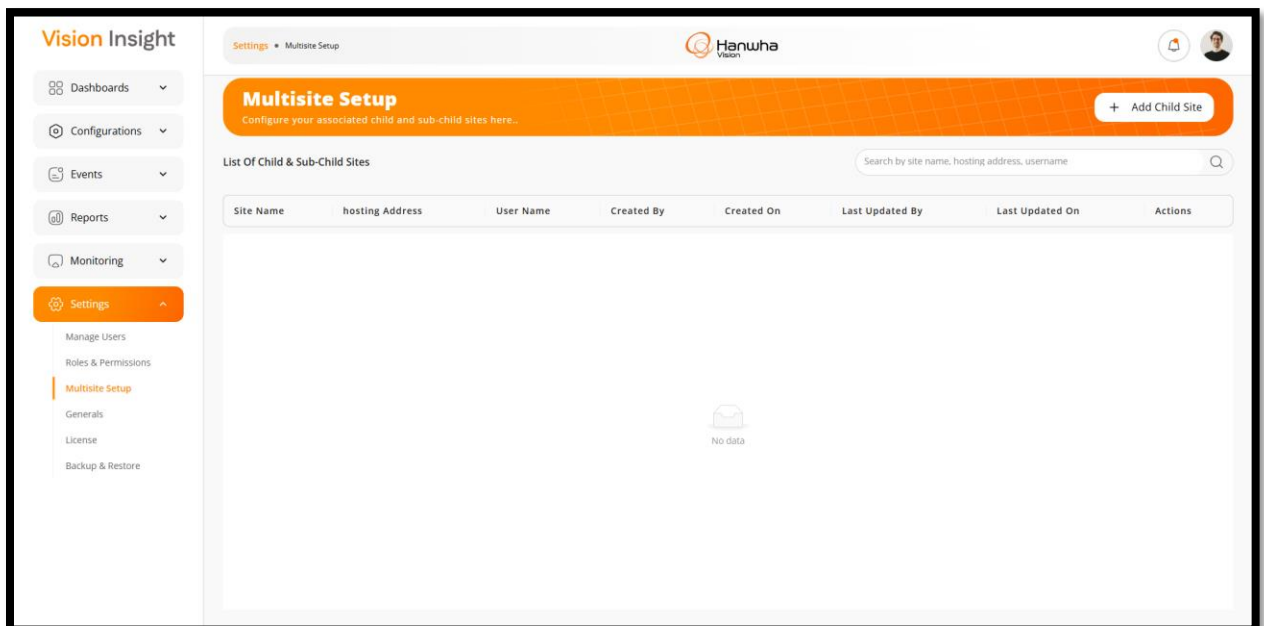


6) Support for managing multiple sites

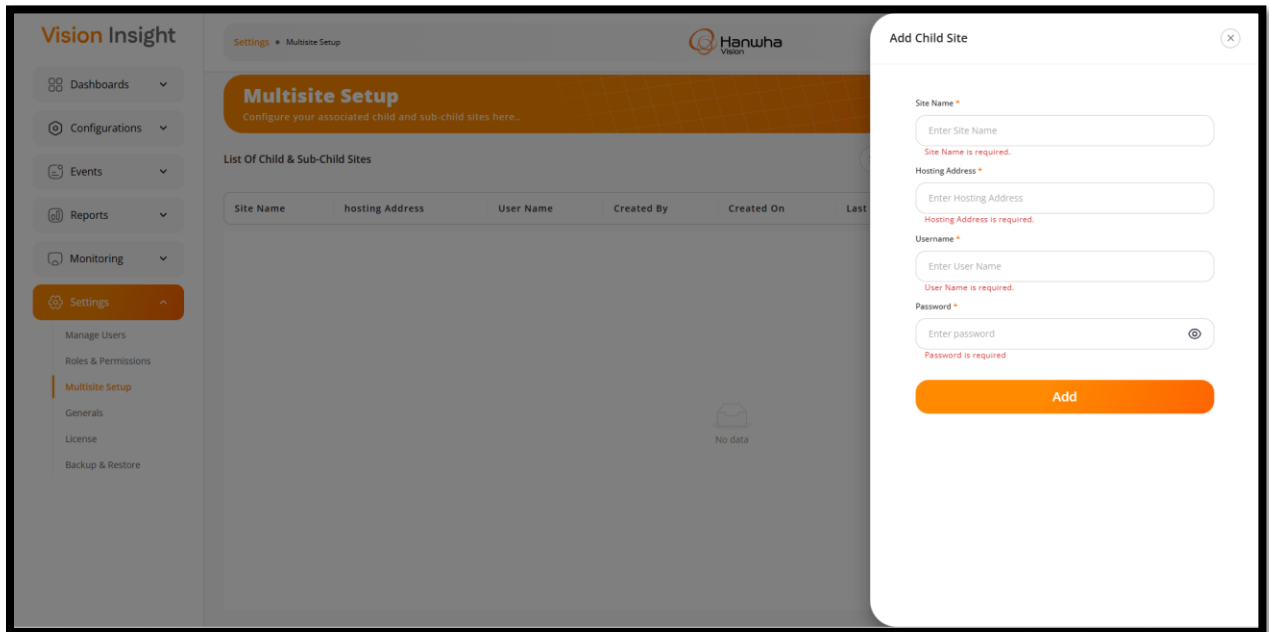
The system allows the parent site's authorized / super admin users to manage single or multiple child and sub-child site configurations.

Screenshots:

➔ The screen below will be displayed when there is **no data found** in the system.



- ➔ The screen below represents the form for creating a child site, allowing users to enter relevant details and submit the information for successful child site creation.
- ➔ The system will perform all necessary validations when adding a new child site to ensure that the entered information is valid and accurate.

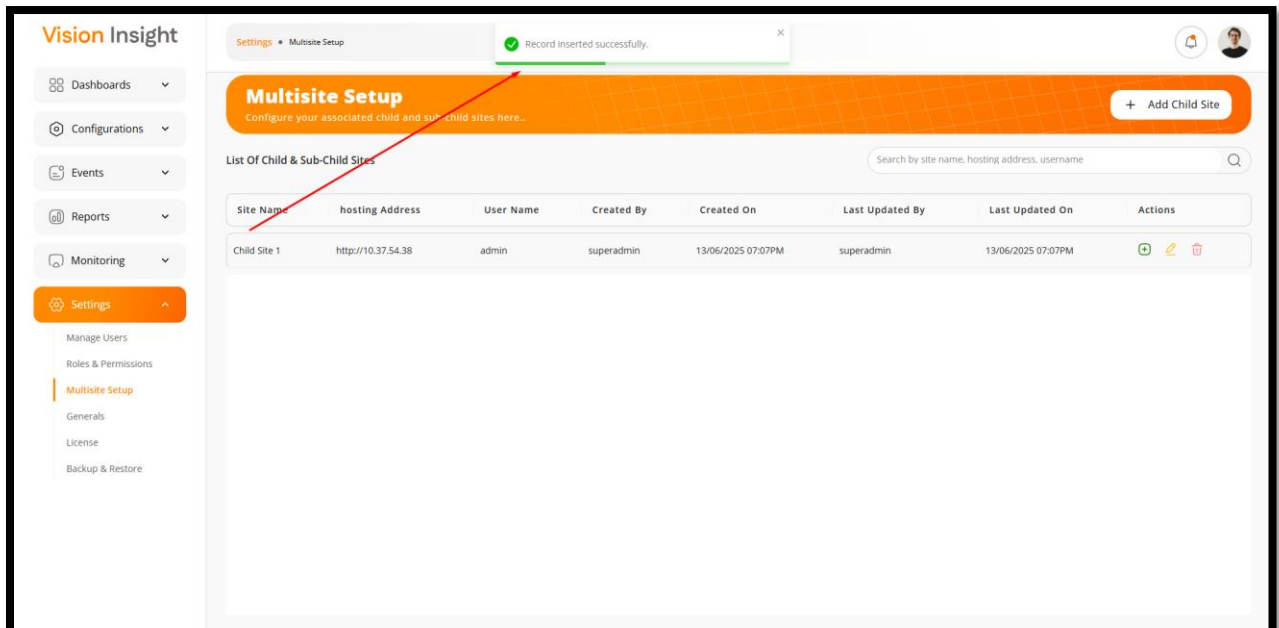


The screenshot shows the 'Multisite Setup' page in the Vision Insight application. A modal form titled 'Add Child Site' is open, allowing users to enter details for a new child site. The form includes the following fields:

- Site Name ***: A text input field with a red error message 'Site Name is required.'
- Hosting Address ***: A text input field with a red error message 'Hosting Address is required.'
- Username ***: A text input field with a red error message 'User Name is required.'
- Password ***: A password input field with a red error message 'Password is required.'

An orange 'Add' button is located at the bottom of the modal. The background shows the 'Multisite Setup' page with a table titled 'List Of Child & Sub-Child Sites' which currently displays 'No data'.

- ➔ The system will display an appropriate success or failure message after the child site is successfully created (or if the creation fails) in the system.

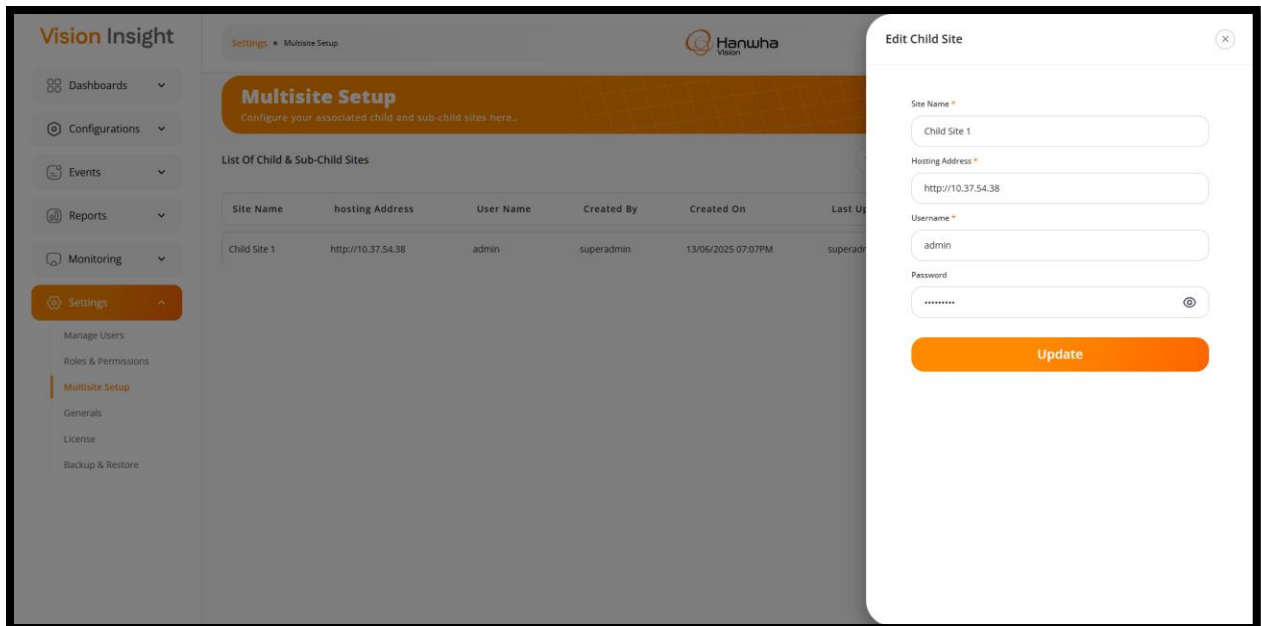


The screenshot shows the 'Multisite Setup' page after a successful record insertion. A green success message 'Record inserted successfully.' is displayed at the top. The 'Add Child Site' button is now '+ Add Child Site'. The table 'List Of Child & Sub-Child Sites' now contains one entry:

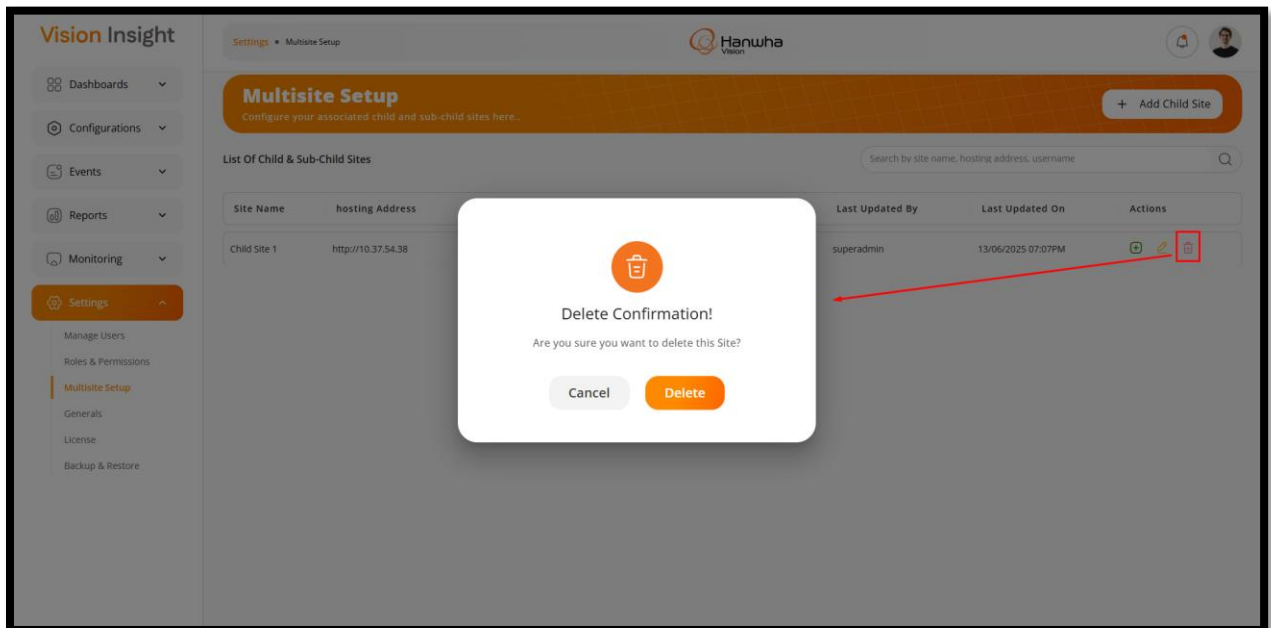
Site Name	hosting Address	User Name	Created By	Created On	Last Updated By	Last Updated On	Actions
Child Site 1	http://10.37.54.38	admin	superadmin	13/06/2025 07:07PM	superadmin	13/06/2025 07:07PM	  

A red arrow points from the success message to the table.

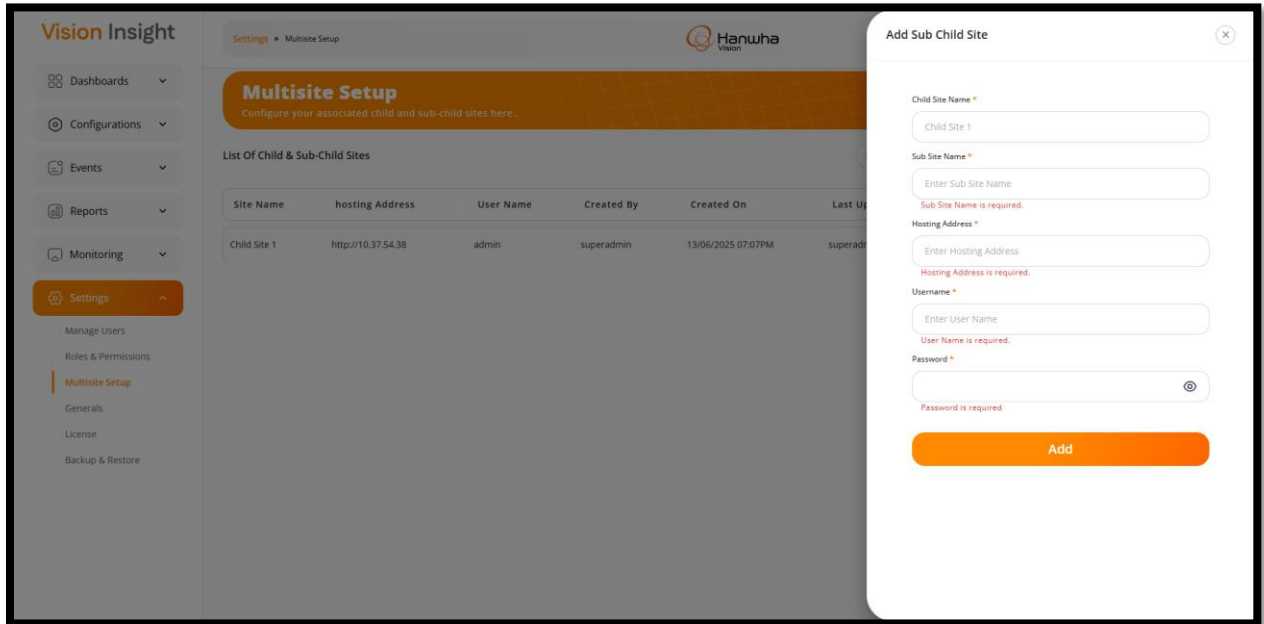
➔ The screen below illustrates the form used for **editing existing child site information**.



➔ The system will **prompt a confirmation message** when the user attempts to **delete a specific child site** from the list.



- ➔ The screen below represents the form for creating a sub-child site, allowing users to enter relevant details and submit the information for successful sub-child site creation.
- ➔ The system will perform all necessary validations when adding a new sub-child site to ensure that the entered information is valid and accurate.

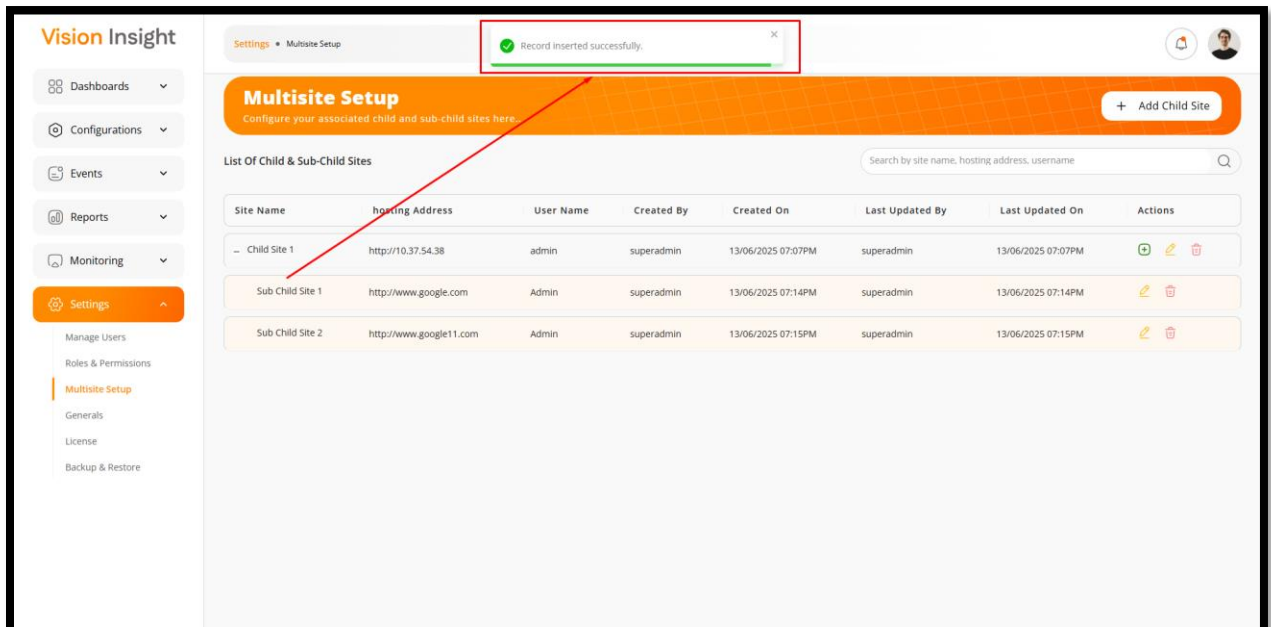


The screenshot shows the 'Add Sub Child Site' modal form. It includes the following fields and validation messages:

- Child Site Name ***: Field contains 'Child Site 1'.
- Sub Site Name ***: Field contains 'Enter Sub Site Name'. Validation message: 'Sub Site Name is required.'
- Hosting Address ***: Field contains 'Enter Hosting Address'. Validation message: 'Hosting Address is required.'
- Username ***: Field contains 'Enter User Name'. Validation message: 'User Name is required.'
- Password ***: Field contains 'Enter Password'. Validation message: 'Password is required.'

An orange 'Add' button is at the bottom right of the modal.

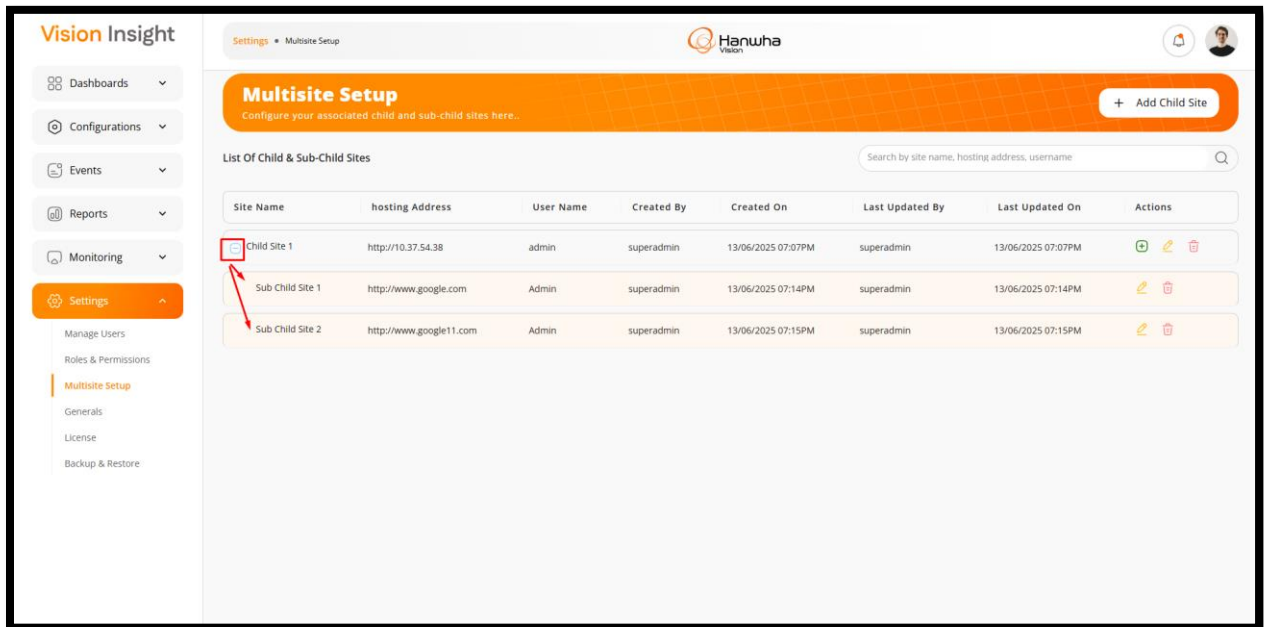
- ➔ The system will display an appropriate success or failure message after the sub-child site is successfully created (or if the creation fails) in the system.



The screenshot shows the 'Multisite Setup' page with a success message: 'Record inserted successfully.' A red arrow points from this message to the 'Sub Child Site 1' row in the table below.

Site Name	hosting Address	User Name	Created By	Created On	Last Updated By	Last Updated On	Actions
Child Site 1	http://10.37.54.38	admin	superadmin	13/06/2025 07:07PM	superadmin	13/06/2025 07:07PM	  
Sub Child Site 1	http://www.google.com	Admin	superadmin	13/06/2025 07:14PM	superadmin	13/06/2025 07:14PM	 
Sub Child Site 2	http://www.google11.com	Admin	superadmin	13/06/2025 07:15PM	superadmin	13/06/2025 07:15PM	 

➔ The **highlighted button below** will appear when **at least one sub-child site exists** under a given **child site**. Users can **expand the list of sub-child sites** by clicking this button.



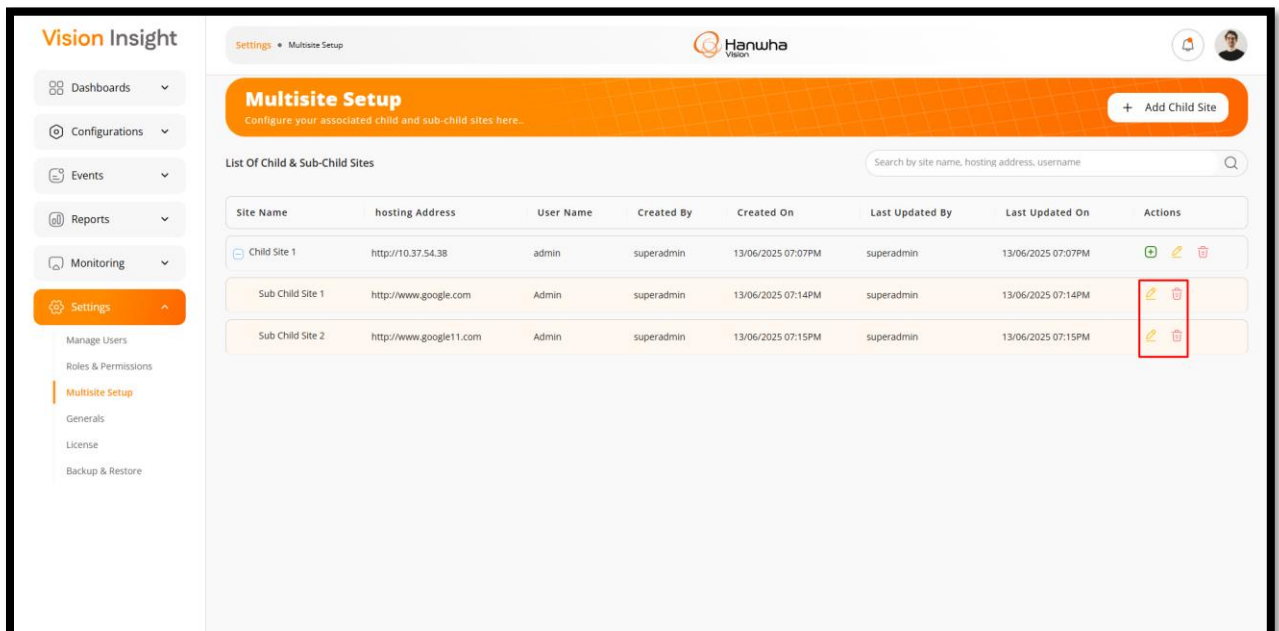
Vision Insight | Settings • Multisite Setup | Hanwha Vision

Multisite Setup
Configure your associated child and sub-child sites here... [+ Add Child Site](#)

Search by site name, hosting address, username

Site Name	hosting Address	User Name	Created By	Created On	Last Updated By	Last Updated On	Actions
Child Site 1	http://10.37.54.38	admin	superadmin	13/06/2025 07:07PM	superadmin	13/06/2025 07:07PM	  
Sub Child Site 1	http://www.google.com	Admin	superadmin	13/06/2025 07:14PM	superadmin	13/06/2025 07:14PM	 
Sub Child Site 2	http://www.google11.com	Admin	superadmin	13/06/2025 07:15PM	superadmin	13/06/2025 07:15PM	 

➔ Users will be able to **edit or delete a sub-child site** by clicking on the **highlighted Edit or Delete buttons** shown below.



Vision Insight | Settings • Multisite Setup | Hanwha Vision

Multisite Setup
Configure your associated child and sub-child sites here... [+ Add Child Site](#)

Search by site name, hosting address, username

Site Name	hosting Address	User Name	Created By	Created On	Last Updated By	Last Updated On	Actions
Child Site 1	http://10.37.54.38	admin	superadmin	13/06/2025 07:07PM	superadmin	13/06/2025 07:07PM	  
Sub Child Site 1	http://www.google.com	Admin	superadmin	13/06/2025 07:14PM	superadmin	13/06/2025 07:14PM	 
Sub Child Site 2	http://www.google11.com	Admin	superadmin	13/06/2025 07:15PM	superadmin	13/06/2025 07:15PM	 

2.4 Data Ingestion (*Partial*)

The system includes a feature that automatically makes specific API calls in the background at predefined time intervals to ingest data into the database. The ingested data is then displayed on the BI Dashboard Web Application.

Since this is a fully automated background process with no user interaction, no permission management is required or handled by the system for this feature.

IMPORTANT NOTE:

Initially, it was planned to deliver this component **partially under Milestone 3**. However, due to changes in the execution plan and a more **comprehensive flow of requirement gathering**, SPEC INDIA successfully developed **all the necessary cron jobs** required for **replicating data from the camera devices**. These background processes will play a **crucial role** in supporting the development of associated **features and functionalities** within the upcoming **BI Dashboard Web Application**.

Screenshots:

- ➔ **Data ingestion** is managed through a set of defined **cron jobs/schedulers** that execute at specific intervals to ensure timely and automated data processing.
- ➔ We have implemented the following **cron jobs** in the system to **replicate data from the camera devices**, enabling consistent and reliable synchronization between the devices and the platform.

Sr.No	Job name	Job Class	Description	Cron
1	DeviceStatus	CheckDeviceStatusJob	Fetch device status and save it in the database	0/30 * * * *
2	PeopleCountJob	PeopleCountJob	Fetch people counting data and save it in the database	0/30 * * * *
3	VehicleCountJob	VehicleCountJob	Fetch vehicle counting data and save it in the database	0/30 * * * *
4	ShoppingCartCount	ShoppingCartCountJob	Fetch shopping cart counting data and save it in the database	0/30 * * * *
5	ForkliftCount	ForkliftCountJob	Fetch forklift counting data and save it in the database	0/30 * * * *
6	MultiLaneVehicleCount	MultiLaneVehicleCountJob	Fetch multilane vehicle counting data and save it in the database	0/30 * * * *
7	ResetCamera	ResetCamera	Reset camera as per configuration time	0/30 * * * *
8	HeatMap	HeatMapJob	Fetch Heat Map data and save it in the database	59 10 * * *
9	OperationalTiming	ManageOperationalTimingJob	It will reset device based on the operational time	0 */5 * * *

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